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A mathematical game for topology optimization of cooling systems

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We propose a topology optimization-based method for optimal design of cooling systems in the form of a mathematical game between two players trying to reach a compromise between limiting the amount of a cooling medium used and obtaining low temperatures in the design domain. The flow of the cooling medium is governed by a Stokes–Brinkman flow model with penalty, while the temperature is governed by a stationary convection–diffusion problem whose solution is approximated using a finite element method with consistent stabilization. Existence of solution for the continuum problems and finite element convergence are shown. The idea and performance of the proposed design method are illustrated by numerical examples based on a problem-setting inspired by an industrial design problem for a gas turbine part. The method exhibits good convergence and is able to generate meaningful design concepts representing various levels of compromise between limited use of cooling medium and low temperatures.

1 | INTRODUCTION

Many products, ranging from phones and computers to jet engines and industrial gas turbines, rely on efficient cooling to function properly. Design of such products requires, as a minimum, taking into account the physics of temperature and fluid flow and design criteria associated with the respective physical domain. Such design criteria are potentially conflicting: on the one hand, it is desirable to keep temperatures low, but on the other hand, one may wish to reduce fan noise, energy spent on pumping a cooling fluid, or limit the amount of coolant used. Optimal design of cooling systems is thus clearly a challenging multiphysics, multicriteria optimization problem.

In this paper, we consider a method for design of cooling systems using topology optimization (TO), a computer-based tool for automatic, optimal design [6, 12]. Existing literature on TO in multiphysics problems include articles on linear [20] and nonlinear [10] thermo-elasticity; design of electro-thermo-mechanical systems [49]; coupled thermal-fluid problems [17, 24, 28, 41], for example, in design of heat sinks [3, 52, 62] or heat exchangers [22]; problems with fluid–structure interaction [17, 32, 35, 40, 58]; and combined flow and mass transport for battery design [60].

The method for design of cooling systems proposed herein is based on a physics model in the form of a coupled stationary thermal-fluid boundary value problem with flow in solid regions penalized using a design-dependent Brinkman term, an idea due to Borrvall and Petersson [8], which is now extensively used in the field [2]. Since its inception in 2003 [8], research on TO in flow problems has gone in two directions: on the one hand, towards more advanced flow models, including Navier–Stokes (N-S) with turbulence models [14, 48, 61]; and on the other hand towards simpler models such as Darcy flow [1, 4, 28, 42, 63]. In the industrial problem for design of internal cooling channels on which our numerical examples are loosely based (see Figure 1 below), we expect designs with many small channels. For such designs, simple flow models

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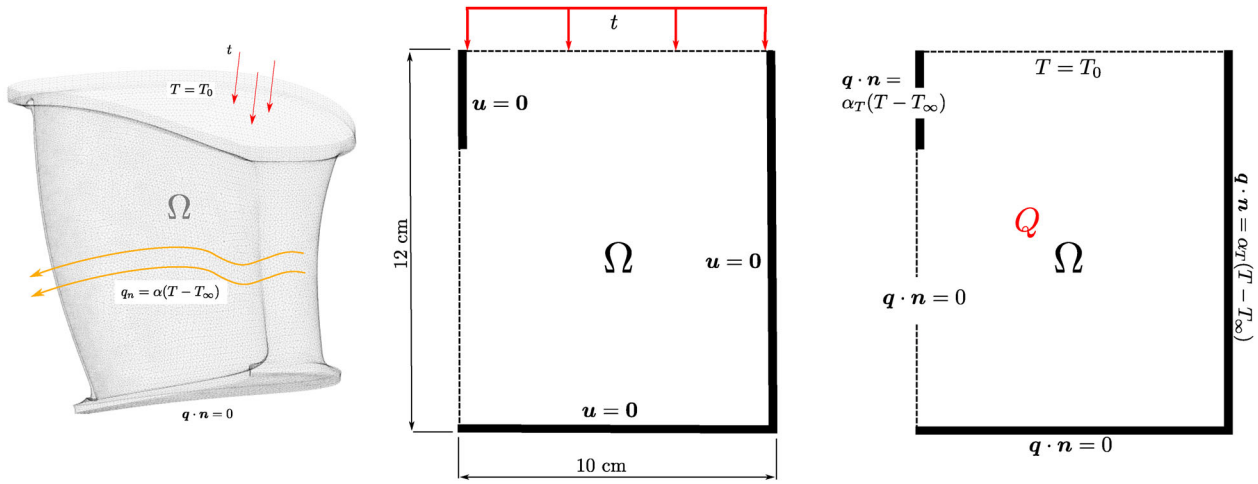


FIGURE 1 Setup for the 2D examples. The left panel shows a quite realistic geometry including thermal boundary conditions and an input pressure (traction) t on the top. The middle and right plots show 2D versions of the setup with boundary conditions and loads for the fluid (left) and the temperature (right) problem. The domain thickness (into the paper) is 3 cm. The thick solid lines indicate nondesignable outer parts. The distributed heat source Q is introduced to model convection on the sides.

have been shown to work well [28] and we, therefore, investigate the use of a Stokes flow model with the aim of achieving meaningful conceptual designs at low-to-moderate computational cost (recent work on Stokes flow in TO include Refs. [31, 37, 38, 56]). Gersborg-Hansen et al. [18], fig. 3] demonstrated that using N-S rather than Stokes flow can lead to substantially different designs, but whether such differences are seen will of course depend on the optimization problem formulation and the Reynolds number. The authors of Ref. [18] sought to minimize viscous drag; here we consider an objective more related to maximizing it, and we did not observe significant differences in the design when optimizing using N-S flow (see Figure 6 below). Nevertheless, more detailed investigations on the use of advanced flow models for our problem setup is an interesting direction for future research.

For the flow problem, we use a penalty approach [36] with only the velocity as an unknown field. Compared to a mixed formulation including also the hydrostatic pressure as an unknown, this leads to a smaller and positive definite stiffness matrix. A drawback of the penalty method is that it adds to the ill-conditioning of the stiffness matrix, thus degrading the performance of iterative linear solvers. For 2D and moderate-sized 3D problems however, the penalty approach with a direct linear solver based on Cholesky factorization is very efficient and robust. An advantage of using a direct solver is the possibility to reuse the factor of the stiffness matrix from the state problem when computing derivatives. This advantage increases with the number of flow-dependent functionals in the problem. In the problem treated herein, there is one state and one adjoint flow problem, so an iterative solver, applied to a larger indefinite mixed formulation, would essentially have to be twice as fast in solving each of these problems to be competitive with the penalty approach. We also remark that the highly accurate solution provided by a direct solver may lead to more accurate derivatives compared to an iterative solver, which may be quite costly if forced to run to the same level of accuracy as a direct solver. A numerical comparison with a mixed (stable) formulation is provided at the end of Section 5.2 and shows no significant differences in terms of optimized design compared to the penalty approach.

A side effect of using a method which allows some compressibility is that, even with appropriate boundary conditions, the stiffness matrix in our temperature problem is not automatically positive definite. We show however that if the penalty parameter is large enough, the bilinear form will be coercive, and, therefore, the stiffness matrix of a conforming finite element (FE) method will indeed be positive definite. This condition is not restrictive, because in practise, a fairly large value of the penalty parameter is needed anyway to ensure that the flow is indeed nearly incompressible; otherwise one typically obtains nonsensical designs without channels (see e.g., fig. 15a in Ref. [25]). The fact that we do not assume complete incompressibility in the continuum problem also enables us to show convergence of solutions of sequences of FE discretized versions of our design problem in Section 4.1 without requiring the penalty parameter to tend to infinity, which would cause numerical difficulties. While FE convergence proofs can be found for elasticity [7, 39] and flow problems [8, 37], we are not aware of any such proofs for the type of thermal-fluid problem treated herein.

In the design-problem treated herein, we seek to simultaneously minimize the average temperature in the solid material and the amount of fluid used for cooling. This leads to a mathematical game [5], or a multicriteria optimization problem

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[33], which is cast as a standard single-criterion optimization problem to minimize a weighted sum of the two objectives. An issue that can arise in such problems is the different scaling of the objectives, and to address this, we use a numerical scaling scheme described in Ref. [19] specialized to our case in Section 3.1 below. The result is a well-behaved optimization problem in which the weights of the two objectives can be controlled in an intuitive way (e.g., small changes of the weights do not lead to wildly different designs), enabling a designer to easily generate different concepts in the range from from designs optimized for minimum fluid flow to designs optimized only for minimum temperature.

2 | THE STATE PROBLEM AND ITS PARAMETRIZATION

The design domain $\Omega \subset \mathbb{R}^d$, $d = 2$ or 3, is an open, bounded set with Lipschitz boundary Γ and outward normal $\boldsymbol{n}$. We use standard density-based TO [6], so the design is described by a scalar field $\rho \in L^\infty(\Omega)$, which should, in each point, ideally only take on the value 0 (fluid) or 1 (solid). The design is subject to the point-wise bounds $0 \le \rho \le 1$ almost everywhere (a.e.) in Ω and penalization described below is used to achieve close to binary-valued designs. Following Bourdin [9], we use a regularized density $\tilde{\rho}$ defined at each point $\boldsymbol{x} \in \Omega$ as the linear convolution

$$\tilde{\rho}(\rho)(\boldsymbol{x}) = \int_\Omega \rho(\boldsymbol{y})\Psi(\boldsymbol{x},\boldsymbol{y})\,\mathrm{d}\boldsymbol{y}, \tag{1}$$

in which the kernel

$$\psi(\boldsymbol{x},\boldsymbol{y}) = \frac{\max(0, R - ||\boldsymbol{x}-\boldsymbol{y}||)}{\int_\Omega \max(0, R - ||\boldsymbol{x}-\boldsymbol{z}||)\,\mathrm{d}\boldsymbol{z}}. \tag{2}$$

Here R is the filter radius, $||\cdot||$ is the Euclidean norm on $\mathbb{R}^d$, and the normalization ensures that $\rho \in [0,1]$ a.e. implies that $\tilde{\rho}(\rho) \in [0,1]$ everywhere in Ω. The regularization allows us to prove existence of a solution in the continuum-problem (see Appendix A), gives some control of the size of the cooling channels, and prevents numerical issues such as FE mesh-dependent designs and checkerboards associated with low-order FEs [50].

2.1 | The flow problem

We consider nearly incompressible Stokes–Brinkman flow with velocity ($\boldsymbol{u} = \boldsymbol{0}$ on $\Gamma_u \subset \Gamma$) and traction ($\boldsymbol{\sigma n} = \boldsymbol{t}$ on $\Gamma_t = \Gamma \setminus \overline{\Gamma_u}$) boundary conditions. The flow problem reads

$$\text{Find } \boldsymbol{u} \in \boldsymbol{V} \ : \ a_F(\rho;\boldsymbol{u},\boldsymbol{v}) = \ell(\boldsymbol{v}), \quad \forall \boldsymbol{v} \in \boldsymbol{V}, \tag{3}$$

where $\boldsymbol{V} = \{\boldsymbol{v} \in \boldsymbol{H}^1(\Omega) \equiv [H^1(\Omega)]^d \mid \boldsymbol{v} = \boldsymbol{0} \text{ on } \Gamma_u\}$ (here and in the following, bold font is used to denote spaces of vector- or matrix-valued functions). The symmetric, bilinear form is given by

$$a_F(\rho;\boldsymbol{u},\boldsymbol{v}) = \int_\Omega \alpha(\tilde{\rho})\boldsymbol{u}\cdot\boldsymbol{v}\,\mathrm{d}V + \int_\Omega 2\mu\boldsymbol{\varepsilon}(\boldsymbol{u}) : \boldsymbol{\varepsilon}(\boldsymbol{v})\,\mathrm{d}V + \int_\Omega \lambda(\nabla\cdot\boldsymbol{u})(\nabla\cdot\boldsymbol{v})\,\mathrm{d}V, \tag{4}$$

where $\boldsymbol{\varepsilon}(\boldsymbol{u}) = \frac{1}{2}(\nabla\boldsymbol{u} + \nabla\boldsymbol{u}^\top)$, $\mu > 0$ is the viscosity, $\lambda \ge 0$ is the penalty parameter, and the inverse permeability [8]

$$\alpha(\tilde{\rho}) = \overline{\alpha} - (\overline{\alpha} - \underline{\alpha})(1-\tilde{\rho})\frac{1+q}{(1-\tilde{\rho})+q}, \tag{5}$$

where $0 \le \underline{\alpha} \ll \overline{\alpha} < \infty$, so that $\tilde{\rho} = 0$ (fluid) implies $\alpha = \underline{\alpha}$ and $\tilde{\rho} = 1$ (solid material) implies $\alpha = \overline{\alpha}$. The parameter $q > 0$, with larger values giving higher penalization of intermediate density values in (0,1) with an appropriate optimization problem formulation (c.f. [34, 45, 51]). Finally, the load functional in (3) is given by

$$\ell(\boldsymbol{v}) = \int_{\Gamma_t} \boldsymbol{t}\cdot\boldsymbol{v}\,\mathrm{d}A, \tag{6}$$

with traction $\boldsymbol{t} \in \boldsymbol{L}^2(\Gamma_t)$.

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Since $\alpha \geq 0$ for all admissible ρ, $\lambda \geq 0$, and $\mu > 0$, coercivity of the bi-linear form (4) follows from Korn's inequality provided $|\Gamma_u| > 0$. Continuity of ℓ over $\boldsymbol{H}^1(\Omega)$ is established using Cauchy–Schwartz inequality followed by a trace theorem [44, Theorem 6, p. 240]. Existence of a unique solution to (3) then follows from the Lax–Milgram lemma.

Remark 1. In most works on TO in flow problems, the inlet boundary condition is a prescribed velocity at a fixed location. Here, however (see the numerical examples below), we prescribe nonzero traction on a part of the design domain, and the optimization solver can itself choose where to add or remove material in order to create inlets. □

2.2 | The temperature problem

The temperature is governed by a linear, stationary boundary value problem with temperature ($T = T_0$ on Γ_T), heat flux ($\boldsymbol{q} \cdot \boldsymbol{n} = 0$ on Γ_q), and convection ($\boldsymbol{q} \cdot \boldsymbol{n} = \alpha_T(T - T_\infty)$ on Γ_α) boundary conditions on three disjoint boundary parts.

The bilinear form for the temperature problem is taken as

$$a_T(\rho, \boldsymbol{u}; T, \tilde{T}) = \int_\Omega k(\tilde{\rho}) \nabla T \cdot \nabla \tilde{T} \, \mathrm{d}V + \int_\Omega c(\boldsymbol{u} \cdot \nabla T)\tilde{T} \, \mathrm{d}V + \int_{\Gamma_{\alpha_T}} \alpha_T T \tilde{T} \, \mathrm{d}A, \tag{7}$$

where the conductivity $k(\tilde{\rho}) = k_f + (k_s - k_f)g(\tilde{\rho})$, with $k_f > 0$ and $k_s > 0$ being the conductivity of fluid and solid, respectively; and $\boldsymbol{u} = \boldsymbol{u}(\rho)$ is the solution to (3). The function g is chosen according to the RAMP scheme [51], so

$$g(\tilde{\rho}) = \frac{\tilde{\rho}}{1 + q_T(1 - \tilde{\rho})}, \tag{8}$$

where $q_T > 0$. The convection coefficient $c > 0$ is taken to be constant. Note that $\boldsymbol{u} \approx \boldsymbol{0}$ in the solid part of the domain, hence the convection term is negligible. Finally, the heat transfer coefficient $\alpha_T > 0$.

The temperature problem now reads:

$$\text{Find } T \in \mathcal{V} \ : \ a_T(\rho, \boldsymbol{u}; T, \tilde{T}) = \ell_T(\rho, \boldsymbol{u}, \tilde{T}) \equiv \ell_0(\tilde{T}) - a_T(\rho, \boldsymbol{u}; \hat{T}_0, \tilde{T}), \quad \forall \tilde{T} \in \mathcal{V}, \tag{9}$$

where $\mathcal{V} = \{T \in H^1(\Omega) \mid T = 0 \text{ on } \Gamma_T\}$ and $\hat{T}_0 \in H^1(\Omega)$ is such that $\hat{T}_0 = T_0$ on Γ_T (the existence of such a function is ensured under the weak assumption that the boundary data $T_0 \in H^{1/2}(\Gamma_T)$ [59, Theorem 8.8]). The Sobolev embedding $H^1(\Omega) \hookrightarrow L^4(\Omega)$ ensures that $c\tilde{T}(\boldsymbol{u} \cdot \nabla T) \in L^1(\Omega)$ so that the convection term in a_T is well-defined. The load-term ℓ_0 in (3) is due to convection on Γ_α and a distributed heat source $Q \in L^2(\Omega)$ and is given by

$$\ell_0(\tilde{T}) = \int_{\Gamma_{\alpha_T}} \alpha_T T_\infty \tilde{T} \, \mathrm{d}A + \int_\Omega Q\tilde{T} \, \mathrm{d}V, \tag{10}$$

where the surrounding temperature $T_\infty \in L^2(\Gamma_{\alpha_T})$.

Due to the convection term, the bilinear form (7) is not symmetric, nor necessarily coercive. To see the latter we use the Poincaré inequality and $\alpha_T > 0$ to get the estimate (in which c_1 is a positive constant)

$$\begin{aligned} a_T(\rho, \boldsymbol{u}; T, T) &= \int_\Omega k(\tilde{\rho}) \nabla T \cdot \nabla T \, \mathrm{d}V + \int_\Omega c(\boldsymbol{u} \cdot \nabla T) T \, \mathrm{d}V + \int_{\Gamma_{\alpha_T}} \alpha_T T^2 \, \mathrm{d}A \\ &\geq c_1 k_f ||T||^2_{H^1(\Omega)} + \int_\Omega c(\boldsymbol{u} \cdot \nabla T) T \, \mathrm{d}V. \end{aligned} \tag{11}$$

To bound the last term from below, we first use appropriate versions of partial integration and the product rule (cf. [16, Theorem 3.8]) to write it as

$$\int_\Omega c(\boldsymbol{u} \cdot \nabla T) T \, \mathrm{d}V = -\frac{1}{2} \int_\Omega c(\nabla \cdot \boldsymbol{u}) T^2 \, \mathrm{d}V + \frac{1}{2} \int_\Gamma c(\boldsymbol{u} \cdot \boldsymbol{n}) T^2 \, \mathrm{d}A. \tag{12}$$

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Substitution into (11) gives

$$a_T(\rho, \boldsymbol{u}; T, T) \geq c_1 k_f ||T||^2_{H^1(\Omega)} - \frac{1}{2}\int_\Omega c(\nabla \cdot \boldsymbol{u}) T^2 \, \mathrm{d}V + \frac{1}{2}\int_\Gamma c(\boldsymbol{u} \cdot \boldsymbol{n}) T^2 \, \mathrm{d}A. \tag{13}$$

If the flow was incompressible, then the second term on the right would vanish. In the penalty method, the flow is not incompressible, but $||\nabla \cdot \boldsymbol{u}||_{L^2(\Omega)}$ can be made arbitrarily small by choosing the penalty parameter λ large enough [36][1]. The integral over Γ in (13) will be non-negative as long as $\boldsymbol{u} \cdot \boldsymbol{n} < 0$ at a point implies $T = 0$ at that point. In our numerical examples this is satisfied because the temperature is prescribed at the inflow of the domain. It follows that there exists a constant $c_2 > 0$ such that

$$a_T(\rho, \boldsymbol{u}; T, T) \geq c_1 k_f ||T||^2_{H^1(\Omega)} - \frac{1}{2}\int_\Omega c(\nabla \cdot \boldsymbol{u}) T^2 \, \mathrm{d}V \geq c_2 ||T||^2_{H^1(\Omega)} \tag{14}$$

for sufficiently large values of λ. The bilinear form a_T is thus, for every admissible design, coercive (and continuous), and existence of a unique solution to (9) follows, therefore, from the Lax–Milgram lemma.

2.3 | The complete state problem

Given an admissible design ρ, the complete, weakly coupled state problem reads

$$\begin{aligned} &\text{Find } (\boldsymbol{u}, T) \in \boldsymbol{V} \times \mathcal{V} \text{ such that} \\ &a_F(\rho; \boldsymbol{u}, \boldsymbol{v}) = \ell(\boldsymbol{v}), \quad \forall \boldsymbol{v} \in \boldsymbol{V} \\ &a_T(\rho, \boldsymbol{u}; T, \tilde{T}) = \ell_T(\rho, \boldsymbol{u}; \tilde{T}), \quad \forall \tilde{T} \in \mathcal{V}. \end{aligned} \tag{15}$$

3 | THE DESIGN PROBLEM

Our design problem is a mathematical game [5, 20, 57] between two players, each seeking to minimize an objective directly associated with either flow or temperature by controlling the design.

The objective of the first player is to minimize the fluid compliance, which since we have no nonzero prescribed velocities amounts to minimizing

$$\phi_F(\rho) = \frac{1}{2}\ell(\boldsymbol{u}(\rho)), \tag{16}$$

where $\boldsymbol{u}(\rho)$ denotes the unique solution to the flow problem for the given design ρ. Minimizing this quantity essentially amounts to minimizing the ($\boldsymbol{t}$-weighted) average flow speed, and thereby the mass flow, at the inlet.

The objective of the second player is to minimize the average temperature in the solid parts of the design domain, that is

$$\phi_T(\rho) = \frac{1}{V}\int_\Omega \tilde{\rho}(\rho) \hat{T}(\rho, \boldsymbol{u}(\rho)) \, \mathrm{d}V \tag{17}$$

where V is the volume of the design domain and multiplication with $\tilde{\rho}(\rho)$ means that the temperature in the fluid ($\tilde{\rho}(\rho) \approx 0$) is made less important. Here $\hat{T}(\rho, \boldsymbol{u}(\rho)) = T(\rho, \boldsymbol{u}(\rho)) + \hat{T}_0$ in which $T(\rho, \boldsymbol{u}(\rho)$ denotes the unique solution to the temperature problem for the given design ρ.

[1] Theorem 3.1 in that reference gives $||\boldsymbol{u} - \boldsymbol{u}_0||_{H^1(\Omega)} \leq C/\lambda$ where C is a positive constant and $\boldsymbol{u}_0$ solves the original, constrained problem and thus satisfies $\nabla \cdot \boldsymbol{u}_0 = 0$. Then $||\nabla \cdot \boldsymbol{u}||_{L^2(\Omega)} = ||\nabla \cdot \boldsymbol{u} - \nabla \cdot \boldsymbol{u}_0||_{L^2(\Omega)} = ||\nabla \cdot (\boldsymbol{u} - \boldsymbol{u}_0)||_{L^2(\Omega)} \leq ||\boldsymbol{u} - \boldsymbol{u}_0||_{H^1(\Omega)} \leq C/\lambda$.

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The set of admissible designs is defined as

$$\mathcal{H} = \left\{ \rho \in L^\infty(\Omega) \mid 0 \le \rho \le 1 \text{ a.e.}, \int_\Omega \tilde{\rho}(\rho)\, \mathrm{d}V = \gamma V \right\}, \tag{18}$$

where $\gamma \in (0,1)$ is the volume fraction of solid material in the domain. The filtered design is set to $\tilde{\rho}(\rho) = 1$ a.e. in Ω_{fix}, with Ω_{fix} being a fixed, non-designable solid part of Ω.

Remark 2. To make the connection with game theory clearer, one could introduce the set of feasible strategies $U \subset \mathcal{H} \times \mathcal{H}$ defined by $U = \{(\rho_F, \rho_T) \in \mathcal{H} \times \mathcal{H} \mid \rho_F = \rho_T\}$ and the so-called biloss operator F defined by $F(\rho_F, \rho_T) = \{\phi_F(\rho_F), \phi_T(\rho_T)\}$. Using these, our game can be cast in so-called standard strategic form [5]. □

3.1 | Problem formulation

There are several different ways to formulate our game [5, Section 10.2]. We shall consider minimizing a weighted sum of the objectives of the two players, leading here to an optimization problem with only simple, linear constraints. A potential issue, however, is the different scaling of the objectives [19]. To achieve a well-balanced optimization problem in which the weights of the objectives can be chosen in a somewhat predictable manner, we follow [19, p. 93] and scale the objectives based on the so-called Utopia and Nadir points in the objective space (here $\mathbb{R}_{\ge 0} \times \mathbb{R}_{\ge 0}$).

The components of the Utopia point $(z_{FU}, z_{TU}) \subset \mathbb{R}^2$ are the (locally) optimal values of the respective individual problem, that is

$$z_{FU} = \min_{\rho \in \mathcal{H}} \phi_F(\rho) \quad \text{and} \quad z_{TU} = \min_{\rho \in \mathcal{H}} \phi_T(\rho), \tag{19}$$

with solutions ρ_F and ρ_T, respectively. In general, there will not exist a design corresponding to the Utopia point, that is, $\rho_F \ne \rho_T$ (of course, neither problem needs to have a unique solution, so more generally one should say that the respective solution sets are disjoint). Therefore this point is also referred to as the shadow or virtual minimum of the game [5, p. 167].

The Nadir point (z_{FN}, z_{TN}) is in a sense opposite to the Utopia point: component i of the Nadir point should be the worst value obtainable for objective i in the design space. However, finding the worst value is itself a nonconvex optimization problem which may be very difficult to solve to global optimality. Therefore, we take here as an approximation, the Nadir point as

$$(z_{FN}, z_{TN}) = (\max\{\phi_F(\rho_T), \phi_F(\rho_0)\}, \max\{\phi_T(\rho_F), \phi_T(\rho_0)\}), \tag{20}$$

where the function evaluations at the initial design ρ_0 are included because our objectives might not be completely contradictory.

We now introduce the scaled objectives

$$f_F(\rho) = \frac{\phi_F(\rho) - z_{FU}}{z_{FN} - z_{FU}} \quad \text{and} \quad f_T(\rho) = \frac{\phi_T(\rho) - z_{TU}}{z_{TN} - z_{TU}}. \tag{21}$$

Assuming that z_{iU} are globally optimal values, $z_{FN} - z_{FU} > 0$ and $z_{TN} - z_{TU} > 0$. However, even if a z_{iU} is not a globally optimal value, it is rather unlikely that z_{iN} would be a better value, so in practise, it is probably safe to assume that the denominators in (21) will be positive. Thus, with some reservations due to the existence of local minima, and assuming that the Nadir point really is the "worst" point, the scaled objectives will satisfy $0 \le f_i(\rho) \le 1$ for every $\rho \in \mathcal{H}$ when combined into a single problem.

Now introduce weights $w_i \ge 0$, $i = 1, 2$, such that $w_1 + w_2 = 1$. Our design problem then reads

$$\min_{\rho \in \mathcal{H}} \{w_1 f_F(\rho) + w_2 f_T(\rho)\}. \tag{22}$$

Existence of at least one globally optimal solution to this problem is shown in Appendix A. If ρ^* is a global minimizer of this problem for given weights, then by Ref. [5, Proposition 3, p. 297], this design is a Pareto optimum, i.e., there is no

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other design that yields a smaller value of f_F *and* f_T. The converse, that all Pareto optima are minimizers of problem (22) for some combination of weights, is true if f_F and f_T are convex, which is not the case here.

4 | FINITE ELEMENT APPROXIMATION

Let $h = \max_{e=1,\dots,m} h_e$ denote the diameter of the largest element in a mesh with $m = m(h)$ elements (for simplicity, we assume that the same mesh is used for both flow and temperature problems). We consider as $h \to 0$ a sequence of shape regular meshes and associated, dense (in the appropriate sense) sequences of approximation spaces $\{\mathcal{H}_h\}$, $\{\boldsymbol{V}_h\}$, and $\{\mathcal{V}_h\}$. The approximation spaces will be finite-dimensional and closed subspaces of $\mathcal{H}$, $\boldsymbol{V}$, and $\mathcal{V}$, respectively, and consist of element-wise infinitely differentiable functions. We assume for simplicity that $\overline{\Omega} = \cup_{e=1}^{m} \overline{\Omega_e}$, where the closure $\overline{\Omega_e} = \Omega_e \cup \Gamma_e$, with Γ_e being the boundary of the element.

The design ρ in (15) and its regularized version $\tilde{\rho}$ are approximated as element-wise constant functions ρ_h and $\tilde{\rho}_h$ with elemental values ρ_e and $\tilde{\rho}_e$, respectively. Letting $\boldsymbol{x}_e$ denote the centroid of element e and taking

$$\tilde{\rho}_e = \frac{\sum_{f=1}^{m} \rho_f \Psi(\boldsymbol{x}_e, \boldsymbol{x}_f) V_f}{\sum_{f=1}^{m} \Psi(\boldsymbol{x}_e, \boldsymbol{x}_f) V_f}, \tag{23}$$

where V_f is the volume of element f, we get that $\tilde{\rho}_e \in [0,1]$ if $\rho_f \in [0,1]$ for all $f = 1, \dots, m$.

The state variables are approximated using Galerkin methods with reduced integration and stabilization as described next.

The FE version of the flow problem (3) reads

$$\text{Find } \boldsymbol{u}_h \in \boldsymbol{V}_h \ : \ a_{Fh}(\rho_h; \boldsymbol{u}_h, \boldsymbol{v}_h) = \ell_h(\boldsymbol{v}_h), \quad \forall \boldsymbol{v}_h \in \boldsymbol{V}_h, \tag{24}$$

where the terms in a_{Fh} are the same as a_F in Equation (4) except for the penalty term, which is approximated using reduced integration to avoid volumetric locking for large values of λ [23]. The Quad4 or HEX8 elements (used in our numerical examples) with reduced integration give an FEM equivalent to a mixed formulation based on the Q1-Q0 element, which may exhibit hydrostatic pressure instabilities. Based on numerical experience by the authors and many others, as well as theoretic analyses such as Ref. [27] and the one in Section 4.1 below, one can, however, expect the approximation of the velocity to be acceptable. We mention as a side note that a tetrahedral element which satisfies a discrete Korn's inequality and allows for reduced integration is presented in Ref. [21].

To ensure a good approximation of the temperature field without having to use impractically fine meshes, the FE version

$$\text{Find } T \in \mathcal{V}_h \ : \ a_{Th}(\rho_h, \boldsymbol{u}_h; T_h, \tilde{T}_h) = \ell_{Th}(\tilde{T}_h) - a_{Th}(\rho_h, \boldsymbol{u}_h; \hat{T}_{0h}, \tilde{T}_h), \quad \forall \tilde{T}_h \in \mathcal{V}_h \tag{25}$$

of the temperature problem includes Galerkin/least-squares (GaLS) stabilization [15]. Therefore,

$$a_{Th}(\rho_h, \boldsymbol{u}_h; T_h, \tilde{T}_h) = a_T(\rho_h, \boldsymbol{u}_h; T_h, \tilde{T}_h) + \sum_{e=1}^{m} \int_{\Omega_e} \tau(c\boldsymbol{u}_h \cdot \nabla \tilde{T}_h - \nabla \cdot (k(\tilde{\rho}_h) \nabla \tilde{T}_h))(c\boldsymbol{u}_h \cdot \nabla T_h - \nabla \cdot (k(\tilde{\rho}_h) \nabla T_h)) \, \mathrm{d}V \tag{26}$$

and

$$\ell_{Th}(\tilde{T}_h) = \ell_0(\tilde{T}_h) + \sum_{e=1}^{m} \int_{\Omega_e} \tau(c\boldsymbol{u}_h \cdot \nabla \tilde{T}_h - \nabla \cdot (k(\tilde{\rho}_h) \nabla \tilde{T}_h)) Q \, \mathrm{d}V, \tag{27}$$

where $\tau = \tau_h(\rho_h, \boldsymbol{u}_h)$ is a stabilization parameter. This parameter should be sufficiently large to prevent nonphysical oscillations in the temperature field but also tend to zero as h goes to zero so that the exact solution is retrieved in the limit. The choice of stabilization parameter is important and discussed in, for example, Refs. [15, 26, 54]. Following Ref. [15, p. 57],

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this parameter is (for a uniform 3D mesh with each element edge parallel to one of the global coordinate axes) taken as

$$\tau = \frac{h}{2}(\overline{x}cu_x + \overline{y}cu_y + \overline{z}cu_z)/||c\boldsymbol{u}_h||^2 \tag{28}$$

where $\overline{x} = \coth Pe_x - 1/Pe_x$, with the directional Peclet number $Pe_x = cu_x h/k(\tilde{\rho}_h)$, and $\overline{y}$ and $\overline{z}$ are defined similarly. The parameter τ may become undefined if one or more velocity components become identically zero, but we have not encountered this in issue practice.

The FE discretized version of the design problem (22) now reads

$$\min_{\rho_h \in \mathcal{H}_h} \{w_1 f_F(\rho_h) + w_2 f_{Th}(\rho_h)\}, \tag{29}$$

where $\mathcal{H}_h = \{\rho_h \mid \rho_h \text{ element-wise constant}, \rho_e \in [0,1], e = 1, \dots, m, \sum_{e=1}^m \tilde{\rho}_e V_e = \gamma V\}$. Since, as we show in the proofs below, for every $\rho_h \in \mathcal{H}_h$, the bi-linear forms are coercive and continuous and the linear forms continuous over $\boldsymbol{V}_h$ and $\mathcal{V}_h$, respectively, the FE state problems (24) and (25) have unique solutions $\boldsymbol{u}_h = \boldsymbol{u}_h(\rho_h)$ and $T_h = T_h(\rho_h, \boldsymbol{u}_h(\rho_h))$.

The matrix version of the fluid compliance is computed as

$$\phi_F(\rho_h) = \frac{1}{2}\ell(\boldsymbol{u}_h(\rho_h)) = \frac{1}{2}\boldsymbol{f}^\top \boldsymbol{u}(\rho_h), \tag{30}$$

in which $\boldsymbol{u}(\rho_h)$ now denotes the coordinate vector of $\boldsymbol{u}_h(\rho_h)$ relative to the FE basis and

$$\boldsymbol{f} = \sum_{e=1}^m \boldsymbol{C}_e^\top \int_{\Gamma_e \cap \Gamma_t} (\boldsymbol{N}_e^F)^\top \boldsymbol{t}\, \mathrm{d}A, \tag{31}$$

in which the matrix $\boldsymbol{C}_e$ relates element-local to global vectors, and $\boldsymbol{N}_e^F$ is the shape function matrix.

The average temperature is approximated as

$$\phi_{Th}(\rho_h) = \frac{1}{V}\int_\Omega \tilde{\rho}_h(\hat{T}_h + \hat{T}_{0h})\,\mathrm{d}V = \underbrace{\frac{1}{V}\tilde{\rho}_e \sum_{e=1}^m \int_{\Omega_e} \boldsymbol{N}_e\,\mathrm{d}V \boldsymbol{C}_e}_{\boldsymbol{L}(\rho_h)^\top} \boldsymbol{T}(\rho_h, \boldsymbol{u}_h(\rho_h)) = \boldsymbol{L}(\rho_h)^\top \boldsymbol{T}(\rho_h, \boldsymbol{u}_h(\rho_h)), \tag{32}$$

where $\boldsymbol{T} = \boldsymbol{T}(\rho_h, \boldsymbol{u}_h(\rho_h))$ collects the temperature degrees of freedom.

Remark 3. Regarding the stabilization in problem (25), we note that $\Delta T_h = 0$ in the numerical examples because these are based on bilinear or trilinear rectangular elements. Then, since $\tilde{\rho}_h$ is element-wise constant, there holds that

$$\nabla \cdot (k(\tilde{\rho}_h)\nabla T_h)) = k(\tilde{\rho}_h)\Delta T_h = 0, \tag{33}$$

meaning that the GaLS method coincides with the streamline upwind/Petrov Galerkin (SUPG) and the stream-line upwind (SU) methods. For the general case in which these methods do not coincide, we speculate that the GaLS method is preferable over SUPG, which does not guarantee a coercive problem for every $\tau \geq 0$; cf. [43, Proposition 8.4.1]. □

4.1 | Finite element convergence

The purpose of this section is to present and prove Theorem 1. To keep it relatively short and simple, the analysis is carried out assuming—as is the case in our numerical examples—SU-stabilization in the temperature problem; that T_0 is constant on Γ_T so that $\hat{T}_0$ can be taken as a constant function; and that the centroid of each element is located in the interior of the element (which is the case if the elements are convex for example).

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Theorem 1. *Every sequence of solutions $\{\rho_h\}$ to problem (29) has a subsequence converging weakly* in $L^\infty(\Omega)$ to a solution ρ^* to problem (22) as $h \to 0$. The corresponding sequence of filtered densities $\tilde{\rho}_h(\rho_h)$ converges point-wise a.e. to $\tilde{\rho}(\rho^*)$.*

The main difficulty in proving this theorem is to show convergence of the FE-discretized velocity and temperature in Lemmas 2 and 5, respectively, but once that has been established, the proof, given at the end of this section, is quite straightforward. In the proofs, we make use of the point-wise a.e. convergence of the filtered design for a weakly* convergent sequence of element-wise constant densities shown in Appendix B.

Lemma 2. *Let $\{\rho_h\}$, $\rho_h \in \mathcal{H}_h$, be a sequence converging weakly* to some $\rho \in \mathcal{H}$. Then the sequence of velocities $\boldsymbol{u}_h = \boldsymbol{u}_h(\rho_h)$ converges strongly (and thus weakly) in $\boldsymbol{H}^1(\Omega)$ to a solution $\boldsymbol{u} = \boldsymbol{u}(\rho)$ to the flow problem in (15).*

Proof. Let (without relabelling) $\{\rho_h\}$ be an arbitrary subsequence and $\{\boldsymbol{u}_h\}$ the associated sequence of velocities.

Since $\boldsymbol{V}_h \subset \boldsymbol{V}$ and the one-point quadrature approximation $I(\nabla \cdot \boldsymbol{u}_h, \nabla \cdot \boldsymbol{u}_h)$ of the penalty term in (4) is positive, coercivity follows as in the continuous case, with constant c_1. Since $\boldsymbol{V}_h \subset \boldsymbol{V}$, we also have continuity of ℓ, with constant c_2. Combining the coercivity and continuity inequalities gives $||\boldsymbol{u}_h||_{\boldsymbol{H}^1(\Omega)} \leq c_2/c_1$, with c_2/c_1 independent of h. By Theorem 5.14-4 in Ref. [11, p. 300], we may thus extract a weakly convergent subsequence, again denoted $\{\boldsymbol{u}_h\}$, converging to some $\boldsymbol{u}$. Since $\boldsymbol{V}$ is weakly closed, this $\boldsymbol{u}$ belongs in $\boldsymbol{V}$.

To see that the limit $\boldsymbol{u} \in \boldsymbol{V}$ satisfies the flow problem, let $\boldsymbol{v} \in \boldsymbol{V}$ be arbitrary and introduce $\boldsymbol{v}_h = \boldsymbol{v}_h(\boldsymbol{v})$ converging strongly (hence weakly) in $\boldsymbol{H}^1(\Omega)$ to $\boldsymbol{v}$ (the existence of such a function follows from the assumed density of $\{\boldsymbol{V}_h\}$ in $\boldsymbol{V}$). Then, adding and subtracting terms,

$$\begin{aligned} |a_F(\rho; \boldsymbol{u}, \boldsymbol{v}) - \ell(\boldsymbol{v})| &\leq |a_F(\rho; \boldsymbol{u}, \boldsymbol{v}) + a_{Fh}(\rho_h; \boldsymbol{u}_h, \boldsymbol{v}_h) - a_{Fh}(\rho_h; \boldsymbol{u}_h, \boldsymbol{v}_h) + \ell(\boldsymbol{v}_h) - \ell(\boldsymbol{v}_h) + \ell(\boldsymbol{v})| \\ &\leq |a_F(\rho; \boldsymbol{u}, \boldsymbol{v}) - a_{Fh}(\rho_h; \boldsymbol{u}_h, \boldsymbol{v}_h)| + |\ell(\boldsymbol{v}_h) - \ell(\boldsymbol{v})|, \end{aligned} \tag{34}$$

where $\ell(\boldsymbol{v}_h) \to \ell(\boldsymbol{v})$ since ℓ is weakly continuous. The term $a_F(\rho; \boldsymbol{u}, \boldsymbol{v}) - a_{Fh}(\rho_h; \boldsymbol{u}_h, \boldsymbol{v}_h)$ can be split into three parts:

$$\int_\Omega \alpha(\tilde{\rho}) \boldsymbol{u} \cdot \boldsymbol{v} \, \mathrm{d}V - \int_\Omega \alpha(\tilde{\rho}_h) \boldsymbol{u}_h \cdot \boldsymbol{v}_h \, \mathrm{d}V, \tag{35}$$

$$\int_\Omega 2\mu \boldsymbol{\varepsilon}(\boldsymbol{u}) : \boldsymbol{\varepsilon}(\boldsymbol{v}) \, \mathrm{d}V - \int_\Omega 2\mu \boldsymbol{\varepsilon}(\boldsymbol{u}_h) : \boldsymbol{\varepsilon}(\boldsymbol{v}_h) \, \mathrm{d}V, \tag{36}$$

and

$$\lambda \int_\Omega (\nabla \cdot \boldsymbol{u})(\nabla \cdot \boldsymbol{v}) \, \mathrm{d}V - \lambda I(\nabla \cdot \boldsymbol{u}_h, \nabla \cdot \boldsymbol{v}_h). \tag{37}$$

Convergence to zero of the first two terms, (35) and (36), was shown in Ref. [56], so we consider only the last term (omitting the constant λ):

$$\begin{aligned} \left| \int_\Omega (\nabla \cdot \boldsymbol{u})(\nabla \cdot \boldsymbol{v}) \, \mathrm{d}V - I(\nabla \cdot \boldsymbol{u}_h, \nabla \cdot \boldsymbol{v}_h) \right| &\leq \left| \int_\Omega (\nabla \cdot \boldsymbol{u})(\nabla \cdot \boldsymbol{v}) \, \mathrm{d}V - \int_\Omega (\nabla \cdot \boldsymbol{u}_h)(\nabla \cdot \boldsymbol{v}_h) \, \mathrm{d}V \right| \\ &\quad + \left| \int_\Omega (\nabla \cdot \boldsymbol{u}_h)(\nabla \cdot \boldsymbol{v}_h) \, \mathrm{d}V - I(\nabla \cdot \boldsymbol{u}_h, \nabla \cdot \boldsymbol{v}_h) \right|. \end{aligned} \tag{38}$$

Here the first term tends to zero by similar arguments as for (36). As for the second term, call it R, let $\varepsilon > 0$ be arbitrary, and write, with $\phi_h = (\nabla \cdot \boldsymbol{u}_h)(\nabla \cdot \boldsymbol{v}_h)$,

$$R \leq \sum_{e=1}^m \left| \int_{\Omega_e} \phi_h \, \mathrm{d}V - \phi_h(\boldsymbol{x}_e) V_e \right| = \sum_{e=1}^m |\phi_h(\boldsymbol{\xi}_e) - \phi_h(\boldsymbol{x}_e)| V_e, \tag{39}$$

where the centroid $\boldsymbol{x}_e \in \Omega_e$ by assumption. Here we have used the integral mean value theorem, so $\boldsymbol{\xi}_e \in \Omega_e$. Since ϕ_h is continuous on each element and $||\boldsymbol{x}_e - \boldsymbol{\xi}_e|| \leq h$, we can choose h small enough that $|\phi_h(\boldsymbol{\xi}_e) - \phi_h(\boldsymbol{x}_e)| < \varepsilon/V$ for all $e = 1, \ldots, m$, whence (39) gives $R < \sum_{e=1}^m \varepsilon V_e / V = \varepsilon$. Returning to (34), we see that both terms in the last line tend to zero,

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so that $|a_F(\rho;\boldsymbol{u},\boldsymbol{v})-\ell(\boldsymbol{v})|\to 0$. Since the flow problem has a unique solution, and our original subsequence was arbitrary it follows that the entire sequence $\{\boldsymbol{u}_h\}$ converges weakly to $\boldsymbol{u}$.

To show strong convergence, we make use of coercivity (with constant c_1) to get

$$\begin{aligned} c_1||\boldsymbol{u}_h-\boldsymbol{u}||_{\boldsymbol{H}^1(\Omega)} \le a_F(\rho_h;\boldsymbol{u}_h-\boldsymbol{u},\boldsymbol{u}_h-\boldsymbol{u}) &= a_F(\rho_h;\boldsymbol{u}_h-\boldsymbol{u},-\boldsymbol{u})+a_F(\rho_h;\boldsymbol{u}_h,\boldsymbol{u}_h)-a_F(\rho_h;\boldsymbol{u},\boldsymbol{u}_h) \\ &= a_F(\rho_h;\boldsymbol{u}_h-\boldsymbol{u},-\boldsymbol{u})+\ell(\boldsymbol{u}_h)-a_F(\rho_h;\boldsymbol{u},\boldsymbol{u}_h), \end{aligned} \tag{40}$$

where the first term in the last line converges to zero using similar (and simpler) arguments as was used to prove $a_F(\rho;\boldsymbol{u},\boldsymbol{v})\to a_{Fh}(\rho_h;\boldsymbol{u}_h,\boldsymbol{v}_h)$ in (34); while the two last terms cancel in the limit since $\ell(\boldsymbol{u}_h)$ converges to $\ell(\boldsymbol{u})$ and $a_F(\rho_h;\boldsymbol{u},\boldsymbol{u}_h)$ to $a_F(\rho;\boldsymbol{u},\boldsymbol{u})$. It follows that $||\boldsymbol{u}_h-\boldsymbol{u}||_{\boldsymbol{H}^1(\Omega)}$ tends to zero. □

To prove coercivity of the bilinear form in the temperature problem, for use in Lemma 5 below, we make use of the following result:

Lemma 3. *Let $\{\boldsymbol{u}_h\}$ be a sequence of solutions to (24) converging strongly in $\boldsymbol{H}^1(\Omega)$ to a solution $\boldsymbol{u}$ of the flow problem (3). Then for every $\varepsilon>0$, there are positive constants C and $\underline{h}$ such that $h<\underline{h}$ implies*

$$||\nabla\cdot\boldsymbol{u}_h||_{L^2(\Omega)}<\varepsilon+C/\lambda. \tag{41}$$

Proof. Let $\boldsymbol{u}_0$ denote the solution to the constrained version of the flow problem (3) with $\nabla\cdot\boldsymbol{u}=0$ a.e. enforced. Then since $\nabla\cdot\boldsymbol{u}_0=0$ a.e., we can write

$$||\nabla\cdot\boldsymbol{u}_h||_{L^2(\Omega)}=||\nabla\cdot\boldsymbol{u}_h-\nabla\cdot\boldsymbol{u}_0||_{L^2(\Omega)}\le||\boldsymbol{u}_h-\boldsymbol{u}_0+\boldsymbol{u}-\boldsymbol{u}||_{\boldsymbol{H}^1(\Omega)}\le||\boldsymbol{u}_h-\boldsymbol{u}||_{\boldsymbol{H}^1(\Omega)}+||\boldsymbol{u}_0-\boldsymbol{u}||_{\boldsymbol{H}^1(\Omega)}. \tag{42}$$

The strong convergence $\boldsymbol{u}_h\to\boldsymbol{u}$ implies that the first term can be bounded by ε for h small enough. From Ref, [36, Theorem 3.1], we get $||\boldsymbol{u}-\boldsymbol{u}_0||_{\boldsymbol{H}^1(\Omega)}\le C/\lambda$ where C is a positive constant. This leads to the asserted estimate $||\nabla\cdot\boldsymbol{u}_h||_{L^2(\Omega)}<\varepsilon+C/\lambda$. □

To prove continuity of the linear and bilinear forms in the FE-discretized temperature problem, we need to bound the stabilization term. The following straightforward result is then useful:

Lemma 4. *For the stabilization parameter defined in (28), there holds that*

$$|\tau|\le hd/(2||c\boldsymbol{u}_h||). \tag{43}$$

Proof. For $|x|<\pi$, we have the Taylor expansion $\coth(x)-1/x\approx 1/x+x/3-1/x+O(x^2)\to 0$ around $x=0$. Therefore, $\coth(x)-1/x\in[0,1]$ for all $h\ge 0$. This implies (in 2D; the 3D case is similar)

$$\begin{aligned} |\tau|=\left|\frac{h}{2}(\overline{x}cu_x+\overline{y}cu_y)/||c\boldsymbol{u}_h||^2\right| &\le \frac{h}{2}\big(|cu_x|+|cu_y|\big)/||c\boldsymbol{u}_h||^2 \\ &\le \frac{h}{2}(||c\boldsymbol{u}_h||+||c\boldsymbol{u}_h||)/||c\boldsymbol{u}_h||^2=hd/(2||c\boldsymbol{u}_h||). \end{aligned} \tag{44}$$

□

Given Lemmas 3 and 4, we can now prove weak convergence of the temperature. One can show strong convergence in the same way as was done for the velocities but that is not necessary for the proof of Theorem 1, so we omit it for the sake of brevity.

Lemma 5. *Let $\{\rho_h\}$, $\rho_h\in\mathcal{H}_h$, be a sequence converging weakly* to some $\rho\in\mathcal{H}$. Then if the penalty parameter λ is large enough, the corresponding sequence of temperatures $\{T_h=T_h(\rho_h,\boldsymbol{u}_h(\rho_h))\}$ converges weakly in $H_1(\Omega)$ to $T=T(\rho,\boldsymbol{u}(\rho))$ solving the temperature problem in (15).*

Proof. Let (without relabelling) $\{\rho_h\}$ be an arbitrary subsequence and $\{T_h\}$ the associated sequence of temperatures.

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To show boundedness of the sequence of temperatures, we will use coercivity of the bilinear form in (25). Since $\mathcal{V}_h \subset \mathcal{V}$ and the stabilization term is non-negative, we can use the same arguments as used for Equation (14) to get (here and in the following, c_i is a positive constant whose meaning can change from one line to the next)

$$a_{Th}(\rho_h, \boldsymbol{u}_h; T_h, T_h) \geq c_1 k_f ||T_h||^2_{H^1(\Omega)} - \frac{1}{2}\int_\Omega c(\nabla \cdot \boldsymbol{u}_h) T_h^2 \, \mathrm{d}V. \tag{45}$$

Using Cauchy—Schwartz and the fact that $\mathcal{V}_h$ is finite-dimensional, so that all norms on this space are equivalent, we get

$$\begin{aligned} a_{Th}(\rho_h, \boldsymbol{u}_h; T_h, T_h) &\geq c_1 k_f ||T_h||^2_{H^1(\Omega)} - \frac{cc_2}{2}||\nabla \cdot \boldsymbol{u}_h||_{L^2(\Omega)}||T_h||^2_{H^1(\Omega)} \\ &= \left(c_1 k_f - \frac{cc_2}{2}||\nabla \cdot \boldsymbol{u}_h||_{L^2(\Omega)}\right)||T_h||^2_{H^1(\Omega)} > c_3||T_h||^2_{H^1(\Omega)} \end{aligned} \tag{46}$$

for some $c_3 > 0$ provided $||\nabla \cdot \boldsymbol{u}_h||_{L^2(\Omega)} < 2c_1 k_f/(cc_2)$. Choosing $\varepsilon = 2c_1 k_f/(cc_2) - C/\lambda > 0$, Lemma 3 gives

$$||\nabla \cdot \boldsymbol{u}_h||_{L^2(\Omega)} < 2c_1 k_f/(cc_2) - C/\lambda + C/\lambda = \frac{2c_1 k_f}{cc_2} \tag{47}$$

for all h small enough. This shows that $a_{Th}(\rho_h, \boldsymbol{u}_h; \cdot, \cdot)$ is coercive for h small enough and λ large enough. We remark that the stabilization improves coercivity but this was not used here where we only made use of the fact that the stabilization term is non-negative.

We now establish continuity of the right-hand side $\ell_{Th}(\tilde{T}_h) - a_{Th}(\rho_h, \boldsymbol{u}_h; \hat{T}_0, \tilde{T}_h)$ in Equation (25). Towards this end, we first note that

$$\ell_{Th}(T_h) = \ell_0(T_h) + \sum_{e=1}^{m}\int_{\Omega_e} \tau(c\boldsymbol{u}_h \cdot \nabla T_h) Q \, \mathrm{d}V = \ell_0(T_h) + \int_\Omega \tau(c\boldsymbol{u}_h \cdot \nabla T_h) Q \, \mathrm{d}V. \tag{48}$$

Using Cauchy–Schwartz and a trace theorem [44, Theorem 6, p. 240], we get

$$|\ell_0(T_h)| \leq |\int_{\Gamma_{\alpha_T}} \alpha_T T_\infty T_h \, \mathrm{d}A + \int_\Omega Q T_h \, \mathrm{d}V| \leq c_1 ||T_h||_{H^1(\Omega)} \tag{49}$$

for some $c_1 > 0$. The second term in Equation (48) can be estimated using Equation (43) as

$$\begin{aligned} \int_\Omega \tau(c\boldsymbol{u}_h \cdot \nabla T_h) Q \, \mathrm{d}V &\leq \int_\Omega |\tau| \, |c\boldsymbol{u}_h \cdot \nabla T_h| \, |Q| \, \mathrm{d}V \\ &\leq \int_\Omega \frac{hd}{2} |\boldsymbol{e}_{u_h} \cdot \nabla T_h| \, |Q| \, \mathrm{d}V \leq hc_2 ||T_h||_{H^1(\Omega)}, \end{aligned} \tag{50}$$

where $\boldsymbol{e}_{u_h} = c\boldsymbol{u}_h/||c\boldsymbol{u}_h||$ and $c_2 = (d/2)||Q||_{L^2(\Omega)}$. Combining this with (49) we get

$$\ell_0(T_h) \leq (c_1 + c_2 h)||T_h||_{H^1(\Omega)}. \tag{51}$$

The other term in the load can be written as

$$a_{Th}(\rho_h, \boldsymbol{u}_h; \hat{T}_{0h}, T_h) = a_T(\rho_h, \boldsymbol{u}_h; \hat{T}_{0h}, T_h) + \int_\Omega \tau(c\boldsymbol{u}_h \cdot \nabla T_h)(c\boldsymbol{u}_h \cdot \nabla T_h) \, \mathrm{d}V, \tag{52}$$

in which continuity of the first term on the right is straight forward to establish. To bound the stabilization term we use (43), Cauchy–Schwartz, and norm-equivalence in finite dimension, to get

$$\begin{aligned} \int_\Omega \tau(c\boldsymbol{u}_h \cdot \nabla T_h)(c\boldsymbol{u}_h \cdot \nabla T_h) \, \mathrm{d}V &\leq c_3 hd \int_\Omega |(\boldsymbol{e}_{u_h} \cdot \nabla T_h)(c\boldsymbol{u}_h \cdot \nabla T_h)| \, \mathrm{d}V \\ &\leq c_3 hd ||\boldsymbol{e}_{u_h} \cdot \nabla T_h||_{L^2(\Omega)} ||c\boldsymbol{u}_h \cdot \nabla T_h||_{L^2(\Omega)} \leq hdc_3 c_4 ||\boldsymbol{u}_h||_{\boldsymbol{H}^1(\Omega)} ||T_h||_{H^1(\Omega)}, \end{aligned} \tag{53}$$

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making use of the inequality $x^2 \leq Mx$ if $|x| \leq M$ in the last line to get rid of one $||T_h||_{H^1(\Omega)}$-term. This leads to the estimate

$$a_{Th}(\rho_h, \boldsymbol{u}_h; \hat{T}_{0h}, T_h) \leq \left(c_1 + c_5 h ||\boldsymbol{u}_h||_{\boldsymbol{H}^1(\Omega)}\right) ||T_h||_{H^1(\Omega)}. \tag{54}$$

Combining this with (51) gives $\ell_{Th} \leq c_1 + c_2 h + c_5 h ||\boldsymbol{u}_h||_{\boldsymbol{H}^1(\Omega)} \leq c_6$ since h tends to zero and $\{\boldsymbol{u}_h\}$ is a convergent, hence bounded sequence. Recalling the coercivity inequality (46), we thus get that, for h small enough, $||T_h||_{H^1(\Omega)} \leq c_3/c_6$.

Since $\{T_h\}$ is bounded, we may now (not relabelling) extract a weakly convergent subsequence. Since $\mathcal{V}_h$ is closed in $\mathcal{V}$, it follows that the limit T belongs in $\mathcal{V}$. It remains to verify that the limit satisfies the temperature problem (9).

Let $\tilde{T} \in \mathcal{V}_h$ be arbitrary, and let $\tilde{T}_h = \tilde{T}_h(\tilde{T}) \in \mathcal{V}_h$ converge strongly to $\tilde{T}$. Then

$$\begin{aligned}
&|a_T(\rho, \boldsymbol{u}; T, \tilde{T}) - \ell_0(\tilde{T}) + a_T(\rho, \boldsymbol{u}; \hat{T}_{0h}, \tilde{T})| \\
&\quad = |a_T(\rho, \boldsymbol{u}; T, \tilde{T}) - \ell_0(\tilde{T}) + a_T(\rho, \boldsymbol{u}; \hat{T}_{0h}, \tilde{T}) - a_{Th}(\rho_h, \boldsymbol{u}_h; T_h, \tilde{T}_h) - \ell_{Th}(\tilde{T}_h) \\
&\qquad - a_{Th}(\rho_h, \boldsymbol{u}_h; \hat{T}_{0h}, \tilde{T}_h) + a_{Th}(\rho_h, \boldsymbol{u}_h; T_h, \tilde{T}_h) + \ell_{Th}(\tilde{T}_h) + a_{Th}(\rho_h, \boldsymbol{u}_h; \hat{T}_{0h}, \tilde{T}_h)| \\
&\quad \leq |\ell_{T_h} - \ell_0| + |a_T - a_{Th}| + |a_{T_0} - a_{Th0}| + |a_{Th0} - \ell_{T_h} + a_{T_h}|,
\end{aligned} \tag{55}$$

omitting arguments for brevity in the last line. Here the last term is zero, and the third term is the same as the second expect that T and T_h are replaced by the function $\hat{T}_0$, which is independent on h. We, therefore, only treat the first two terms in detail, starting with

$$\begin{aligned}
|\ell_0(\tilde{T}) - \ell_{Th}(\tilde{T}_h)| &\leq |\ell_0(\tilde{T}) - \ell_0(\tilde{T}_h)| + \left| \int_\Omega \tau(c\boldsymbol{u}_h \cdot \nabla \tilde{T}_h) Q \, \mathrm{d}V \right| \\
&\leq |\ell_0(\tilde{T}) - \ell_0(\tilde{T}_h)| + c_2 h ||\tilde{T}_h||_{H^1(\Omega)},
\end{aligned} \tag{56}$$

where (50) was used for the second inequality. The first term in the last line tends to zero due to the weak convergence of $\tilde{T}_h$ to $\tilde{T}$, and the second since h tends to zero and $||\tilde{T}_h||_{H^1(\Omega)}$ is bounded independently of h. Then

$$|a_T - a_{Th}| \leq |a_T(\rho, \boldsymbol{u}; T, \tilde{T}) - a_T(\rho_h, \boldsymbol{u}_h; T_h, \tilde{T}_h)| + \left| \int_\Omega \tau(c\boldsymbol{u}_h \cdot \nabla T_h)(c\boldsymbol{u}_h \cdot \nabla T_h) \, \mathrm{d}V \right| \tag{57}$$

where the convergence of the first term is handled similarly as in the fluid problem (cf. also Lemma A.1), and the convergence of the last term follows from the estimate (53) and the fact that $\{T_h\}$ is bounded in $H^1(\Omega)$.

We have thus shown that the limit T of our subsequence $\{T_h\}$ satisfies the thermal state problem. Since the original subsequence was arbitrary and, for λ large enough, this problem has a unique solution, it follows that the entire sequence converges. □

Finally, before proving the main theorem, we show that the objective functions in Equation (29) are continuous with respect to the type of convergence just proven:

Lemma 6. *Let $\{(\rho_h, \boldsymbol{u}_h, T_h)\}$, $\rho_h \in \mathcal{H}_h$, $\boldsymbol{u}_h = \boldsymbol{u}_h(\rho_h)$, $T_h = T_h(\rho_h, \boldsymbol{u}_h(\rho_h))$, be a sequence converging weakly* × weakly × weakly to some $(\rho, \boldsymbol{u}(\rho), T(\rho, \boldsymbol{u}(\rho))) \in \mathcal{H} \times \boldsymbol{V} \times V$. Then $f_F(\rho_h) = f_F(\boldsymbol{u}_h(\rho_h)) \to f_F(\boldsymbol{u}(\rho)) = f_F(\rho)$ and $f_{Th}(\rho_h) = f_{Th}(\rho_h, T_h(\rho_h, \boldsymbol{u}_h(\rho))) \to f_T(\rho, T(\rho, \boldsymbol{u}(\rho))) = f_T(\rho)$, respectively.*

Proof. The two functions f_F and f_T are proportional to ϕ_F and ϕ_T, respectively. Therefore, the first part of the lemma is immediate from

$$\phi_F(\rho_h) = \frac{1}{2}\ell(\boldsymbol{u}_h(\rho_h)) \to \frac{1}{2}\ell(\boldsymbol{u}(\rho)), \tag{58}$$

which holds since ℓ is weakly continuous. Then,

$$\phi_{Th}(\rho_h) = \frac{1}{V}\int_\Omega \tilde{\rho}_h(\rho_h) T_h \, \mathrm{d}V = \frac{1}{V}\int_\Omega [\tilde{\rho}_h(\rho_h) - \tilde{\rho}(\rho)] T_h \, \mathrm{d}V + \frac{1}{V}\int_\Omega \tilde{\rho}(\rho) T_h \, \mathrm{d}V, \tag{59}$$

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where the last integral tends to zero because of the weak convergence of $\{T_h\}$ to T. Cauchy–Schwartz inequality then gives the estimate

$$\left|\frac{1}{V}\int_\Omega [\tilde{\rho}_h(\rho_h)-\tilde{\rho}(\rho)]T_h\,\mathrm{d}V\right| \le \frac{1}{V}||\tilde{\rho}_h(\rho_h)-\tilde{\rho}(\rho)||_{L^2(\Omega)}||T_h||_{L^2(\Omega)}. \tag{60}$$

Since $|\tilde{\rho}_h(\rho_h)-\tilde{\rho}(\rho)| \le 1$ and $\tilde{\rho}_h(\rho_h) \to \tilde{\rho}(\rho)$ a.e., we can use the Lebesgue dominated convergence theorem and the fact that $\{T_h\}$ is a bounded sequence to conclude that right-hand side in (60) tends to zero, and thus from (59) that $\phi_{Th}(\rho_h) \to \phi_T(\rho)$. □

We are now ready to give the

Proof of Theorem 1. By definition, a solution ρ_h^* to problem (29) satisfies

$$w_1 f_F(\rho_h^*) + w_2 f_{Th}(\rho_h^*) \le w_1 f_F(\rho_h) + w_2 f_{Th}(\rho_h), \quad \forall \rho_h \in \mathcal{H}_h. \tag{61}$$

For every $\rho \in \mathcal{H}$, there is a function $\rho_h = \rho_h(\rho)$ such that $\rho_h(\rho) \in \mathcal{H}_h$ and $\{\rho_h(\rho)\}$ tends weakly* to $\rho \in \mathcal{H}$ (c.f., e.g., Ref. [47, Proposition 9, p. 129] and note that strong convergence in $L^2(\Omega)$ implies weak* convergence in the present situation). Substitution into inequality (61) gives

$$w_1 f_F(\rho_h^*) + w_2 f_{Th}(\rho_h^*) \le w_1 f_F(\rho_h(\rho)) + w_2 f_{Th}(\rho_h(\rho)), \quad \forall \rho \in \mathcal{H}. \tag{62}$$

Due to the constraint $\rho_h \in [0,1]$ a.e. in Ω, any sequence $\{\rho_h\}$, $\rho_h \in \mathcal{H}_h$, is bounded, hence by Ref. [30, Theorem 12, p. 107], admits a subsequence converging weakly* to some $\rho \in L^\infty(\Omega)$. Since $\mathcal{H}_h \subset \mathcal{H}$ and $\mathcal{H}$ is weakly* closed, it follows that $\rho \in \mathcal{H}$. This implies in particular that we may from the sequence of FE-solutions $\{\rho_h^*\}$ extract a weakly* convergent subsequence converging to some $\rho^* \in \mathcal{H}$ and, according to Lemmas 2 and 5, such that the corresponding sequences of velocities and temperatures converge weakly to some $\boldsymbol{u} = \boldsymbol{u}(\rho^*)$ and $T = T(\rho^*)$, respectively. Recalling the weak* convergence of $\{\rho_h(\rho)\}$ together with Lemmas 2 and 5, and using the continuity from Lemma 6, inequality (62) then gives, in the limit as $h \to 0$,

$$w_1 f_F(\rho^*) + w_2 f_T(\rho^*) \le w_1 f_F(\rho) + w_2 f_T(\rho), \quad \forall \rho \in \mathcal{H}, \tag{63}$$

that is, ρ^* solves problem (22). □

5 | NUMERICAL EXAMPLES

The overall framework for solving problem (29) is implemented in Matlab (R2021b), with some time-consuming parts in the form of C++ MEX files using OpenMP for parallelization. As optimization solver, we use the method of moving asymptotes (MMA) [53], with default settings [53] unless otherwise stated, together with analytical first-order derivatives (Appendix C). The Matlab-function `decomposition` is used for efficient re-use of the numerical factorizations (Cholesky and LU, respectively) of the stiffness matrices $\boldsymbol{K}_F$ and $\boldsymbol{K}_T$ when solving the adjoint problems to compute derivatives. The 2D-examples use uniform meshes with four-noded, bilinear FEs (Quad4) with 2×2 Gauss quadrature [23] for all matrices except for the penalty term in (24), which is evaluated using 1-point (the centroid) Gauss quadrature. In the 3D example, we use eight-noded, tri-linear FEs (HEX8) with $2 \times 2 \times 2$ Gauss quadrature on all matrices expect for the 1-point rule on the penalty term. The amount of solid material in the design domain is limited to 50% by setting $\gamma = 0.5$; the initial guess for the design is in all cases taken as a uniform distribution satisfying the volume constraint; and the filter radius R is set to 1 mm unless stated otherwise. The design plots show the filtered design with values below 0.5 omitted.

The setup for the 2D examples is illustrated in Figure 1. It is inspired by a design problem provided by Siemens Energy AB to design the interior of a so-called guide vane, a wing-like structure used to guide the flow of hot gas inside a gasturbine. The guide vane, which can be fitted into a box of size $10 \times 3 \times 12$ cm, is heated by hot gas flowing over the sides and internal cooling is achieved by pumping compressed air through it. Compressed air flows in at the top and is allowed to exit on the left side.

The viscosity in the flow problem is set to $\mu = 1$ kg/m·s. This is higher than the actual value but results in reasonable flow speeds, and can, at least partly, be motivated as a way to model the effect of turbulence not present in the Stokes model. The magnitude of the traction at the inflow (Figure 1) is set to $4 \cdot 10^5$ Pa. The penalty parameter λ is set to 10^{10} kg/m·s.

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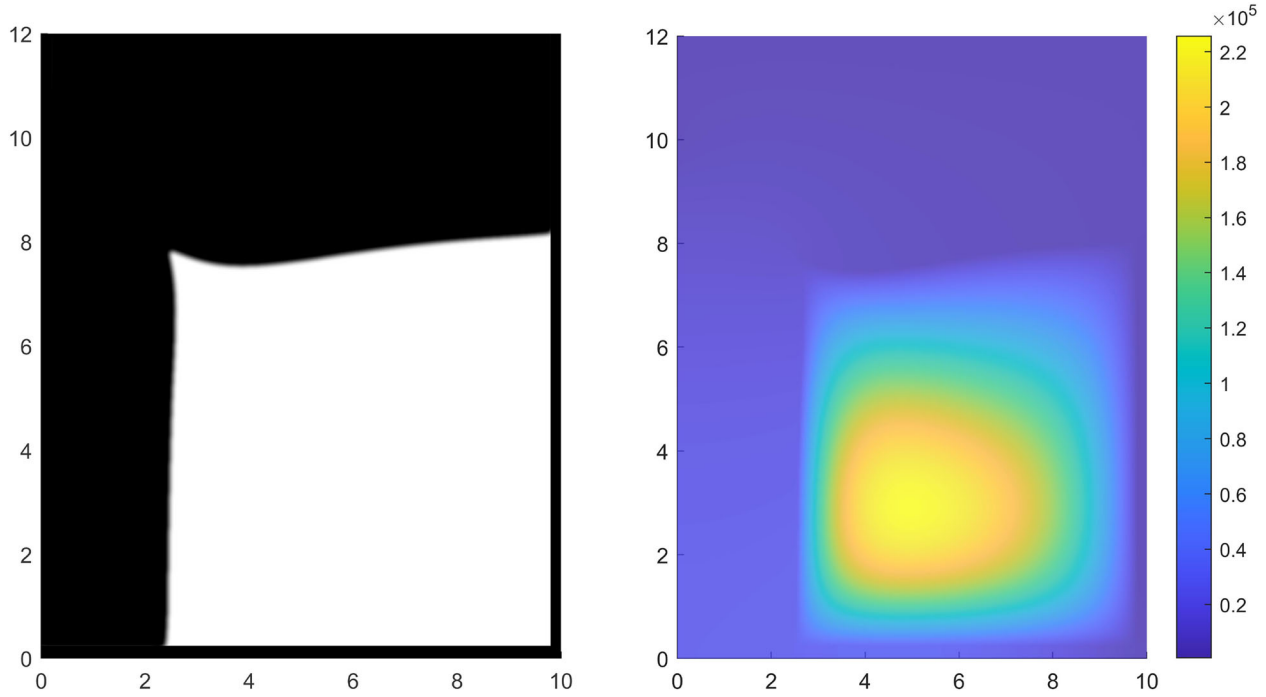


FIGURE 2 Left: Design obtained from minimizing the fluid compliance. Dimensions in centimeter. Right: Temperature [K]

As noted already in Ref. [8], a gradual increase of $\overline{\alpha}$ and q in (5) during the optimization process is typically necessary to obtain reasonable designs. The values chosen are given in each case below. Ideally the choice of when and how much to change these parameters should be done in an adaptive manner, automatically balancing the desires to minimize the objective and to have discrete-valued designs, but the only work we are aware of that describes such a method is Ref. [46]. Since the heuristic procedure of just increasing the values $\overline{\alpha}$ and q after some fixed values works well (and is much used in the literature), we leave development of an adaptive scheme as an idea for future research. In the 2D examples we take $\underline{\alpha} = 10\mu/b^2$ as suggested in Ref. [8] (though our numerical experiments show that for example $\underline{\alpha} = 0$ results in very similar designs), and in 3D, we set $\underline{\alpha} = 0$.

In the temperature problem, we set $T_\infty = 1373$ K, $T_0 = 673$ K, the conductivity of the solid $k_s = 10.2$ W/mK, and the conductivity of air to $k_f = 0.032$ W/mK. The heat transfer coefficient $\alpha_T = 3500$ W/m^{2}K. The convection coefficient $c = \rho_f c_v$ J/K·m^3, where $\rho_f = 5$ [kg/m^3] is the actual density of the cooling medium and $c_v = 1001$ J/K·kg the specific heat. The heat source Q is taken as a constant field with magnitude $20 \cdot 10^6$ W/m^3 acting over the entire design domain. With the dimensions given in Figure 1, this results in a total of 7200 W supplied to the design domain by Q.

In the following, we first study the flow and temperature design problem separately, that is, we look at designs obtained from attempting to solve separately the two problems in (19). Such "no-compromise designs," which represent the best achievable performance in the respective physics domain, are needed to compute the Utopia points in the scalings (21), but may also be useful to a designer when deciding which "compromise designs" are acceptable. Examples of compromise designs obtained with different weights in the complete problem (29) are given in Section 5.3.

5.1 | Pure fluid problem

Figure 2 shows a pure fluid design on a mesh with 800×960 elements (such a fine mesh is not really necessary for this particular problem but useful below). This gives element side lengths of 0.125 mm, whence the set filter radius of 1 mm corresponds to eight such lengths. Solid material has been placed to block inflow at the top of the domain and outflow on the left side, leading to velocities which are very close to zero (but not exactly since $\overline{\alpha} < \infty$) in the entire domain. As can be seen in the temperature plot, the temperature is extremely high in the void region in the lower right corner. This is expected since the thermal conductivity is very low, essentially k_f, and there is no fluid to transport away heat by convection. The design in Figure 2 represents an extreme case which is probably of little practical interest, but is the

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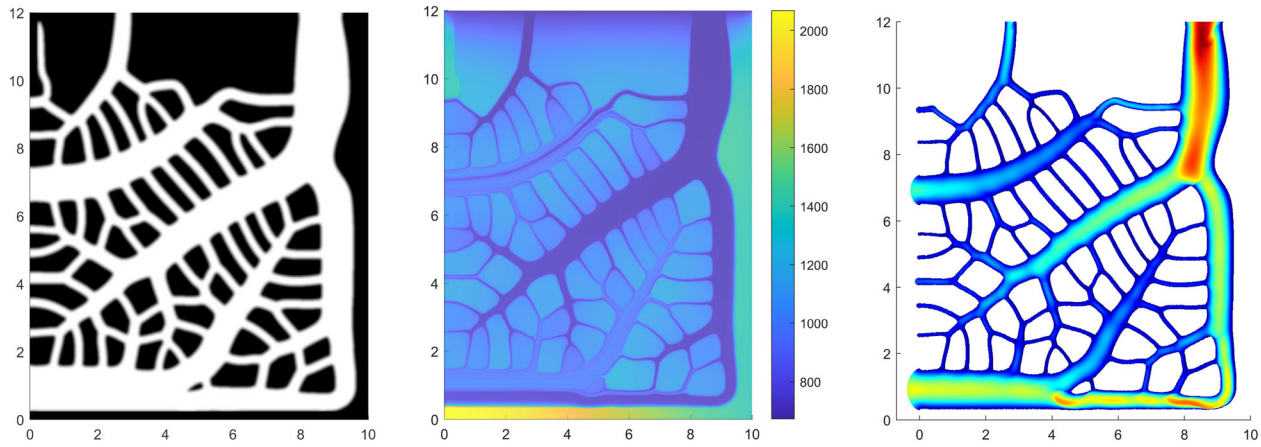


FIGURE 3 Design optimized for temperature only. Left: Design. Middle: Temperature [K]. Right: Flow

best result possible (the design could of course be a local minimum but this appears unlikely) for the fluid player with the given constraints.

5.2 | Pure temperature problem

In the pure temperature problem, the designs are a lot more interesting and complex. Figure 3 shows a design obtained with the same parameters as in the previous section. A network of channels distributed fairly evenly throughout the domain manages to keep the temperature down in large parts of the domain. However, high temperatures can be observed on the right and bottom sides (middle plot). This is not surprising given that the right side is heated both by convection and the volumetric heat source; the bottom side is isolated; and that we minimize an average rather than (some smooth approximation of) the maximum temperature. In the real case, the bottom side is attached to the inner stator and some heat transfer would occur; hence perfect isolation could be replaced by a more realistic boundary condition if desired (and if good data is available). The temperature field has a smooth appearance and it may be noted that the mesh is fine enough that switching off the stabilization has no visible impact on the field. The rightmost plot in Figure 3 illustrates the flow, with dark blue representing the minimum and dark red the maximum speed (roughly 18 m/s).

Figure 4 shows iteration history and designs at the end of each continuation step. As can be seen in the upper right plot, staying at $(\overline{\alpha}, q) = (10^8, 0.01)$ would have given a somewhat difficult-to-interpret design with many intermediate density values, whereas the design at $(10^8, 0.1)$ in the lower left corner might actually be acceptable, especially considering that what we show here should be thought of as conceptual designs.

Figure 5 shows the results when solving the same problem for three different mesh densities. As expected, the design on the finer meshes is refined but qualitatively similar versions of the design on the coarsest mesh.

To assess the impact of using the penalty approach for the flow problem, we have compared designs obtained using a mixed formulation based on the nonconforming mixed P1-P0 element proposed by Kouhia and Stenberg [29], a very simple element which, in contrast to the Crouzeuix–Raviart P1-P0 element for example, can handle traction boundary conditions. The mesh for the flow was obtained from the temperature mesh by dividing each rectangle into two triangles. Figure 6 (left and middle) shows examples of designs obtained with the two different methods. As can be seen, the designs are very similar. We have also solved the problem using the stabilized Q1-Q0 method recently analyzed for TO in flow problems in Ref. [56] and obtained essentially the same design.

Another interesting question is whether switching from Stokes to (stationary) N-S flow has any significant impact on the optimized designs? The right-most plot in Figure 6 shows an optimized design obtained using stationary N-S flow with SUPG-stabilization, with the stabilization parameter defined in Ref. [55, eq. (3.10)], for the convective acceleration. Comparing with the other two designs, we see that they are qualitatively very similar. The N-S equations were solved using the trust-region dogleg version of Newton's method implemented in the Matlab-function `fsolve`, and on average five Newton iterations was needed to reach convergence to high accuracy for a given design.

Figure 7 shows results obtained from a 3D version of the pure temperature problem. In this case, there is no distributed heat source Q; instead, heat is supplied by convection on the sides of the now box-shaped, $10 \times 3 \times 12$ cm design domain

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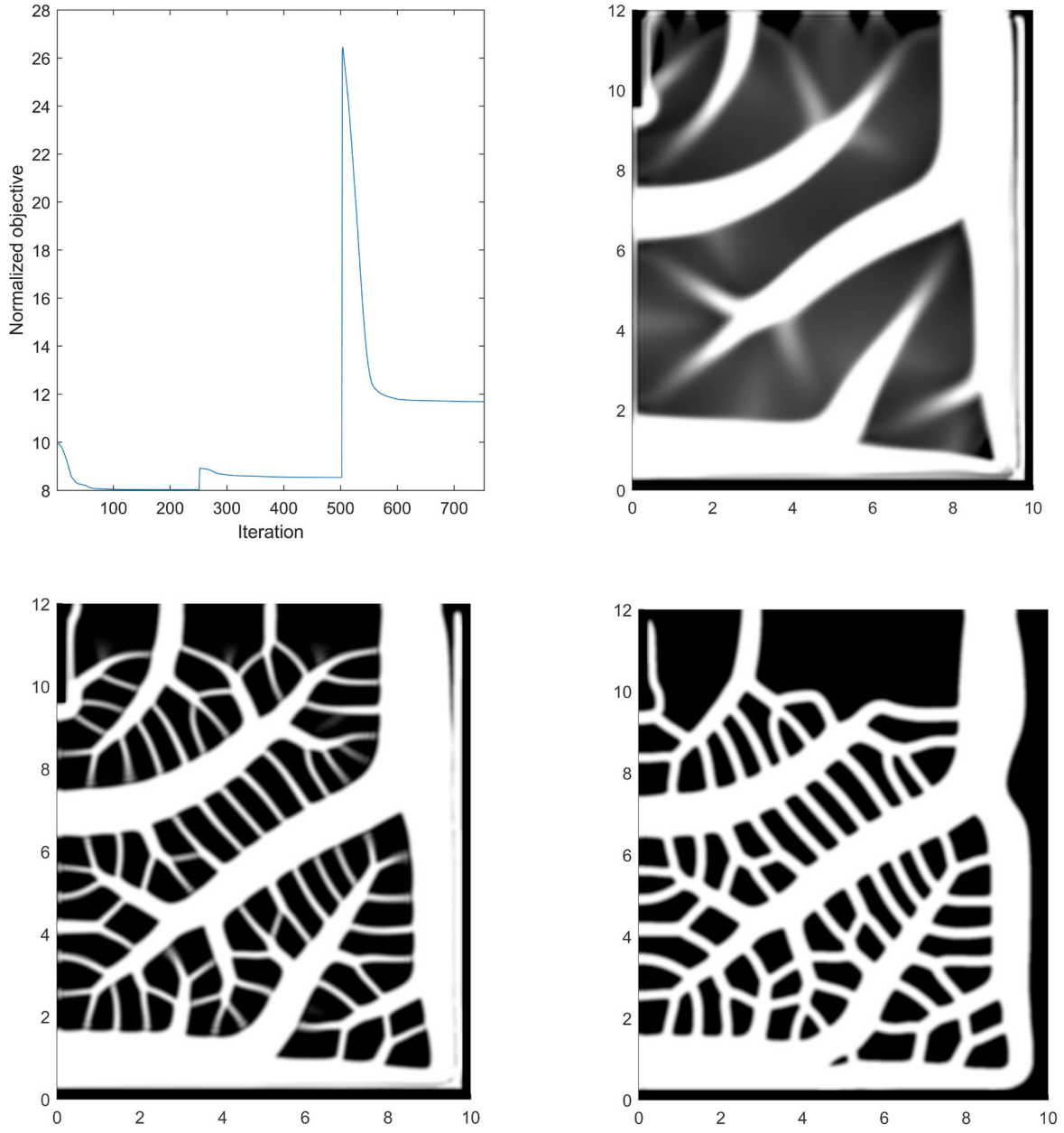


FIGURE 4 Iteration history (the jumps are due to to changes in q and $\overline{\alpha}$ at iteration 250 and 500) and designs at the end of each continuation step. Top right: $(\overline{\alpha}, q) = (10^8, 0.01)$. Bottom left: $(\overline{\alpha}, q) = (10^8, 0.1)$. Bottom right: $(\overline{\alpha}, q) = (10^{10}, 0.1)$.

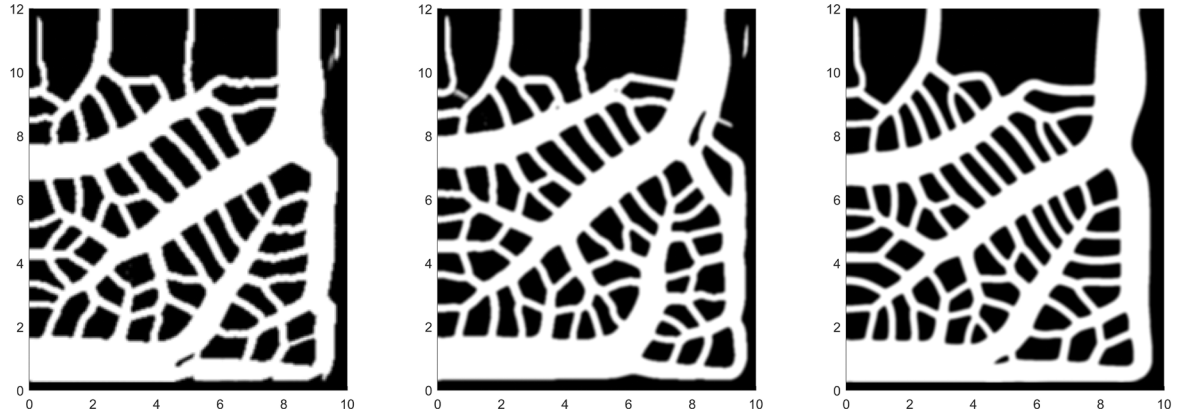


FIGURE 5 Design for three different mesh densities. Left: 200×240. Middle: 400×480. Right: 800×960.

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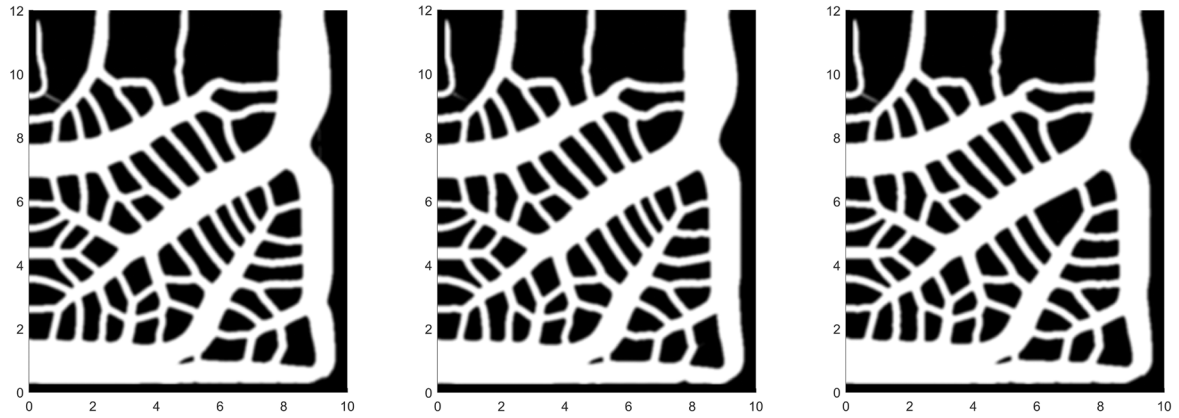


FIGURE 6 Designs using a 400×480 mesh. Left: Design obtained with the Quad4 penalty approach for the flow. Middle: Design obtained with nonconforming mixed P1-P0 elements [29] for the flow. Right: Design obtained using the Quad4 penalty approach and Navier–Stokes flow.

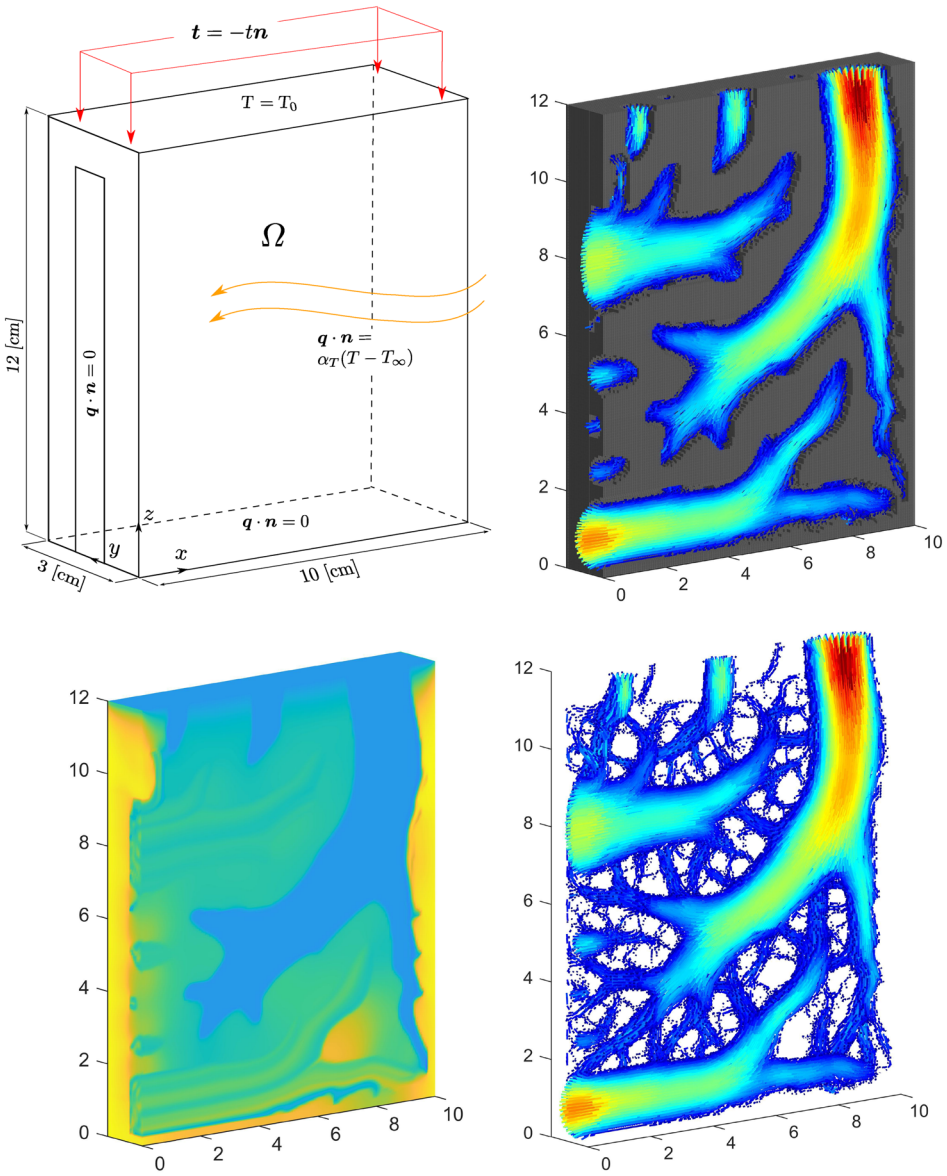


FIGURE 7 3D version of the pure temperature problem. Top left: Setup with boundary conditions. Top right: Design with flow. Bottom left: Flow only. Bottom right: Temperature field.

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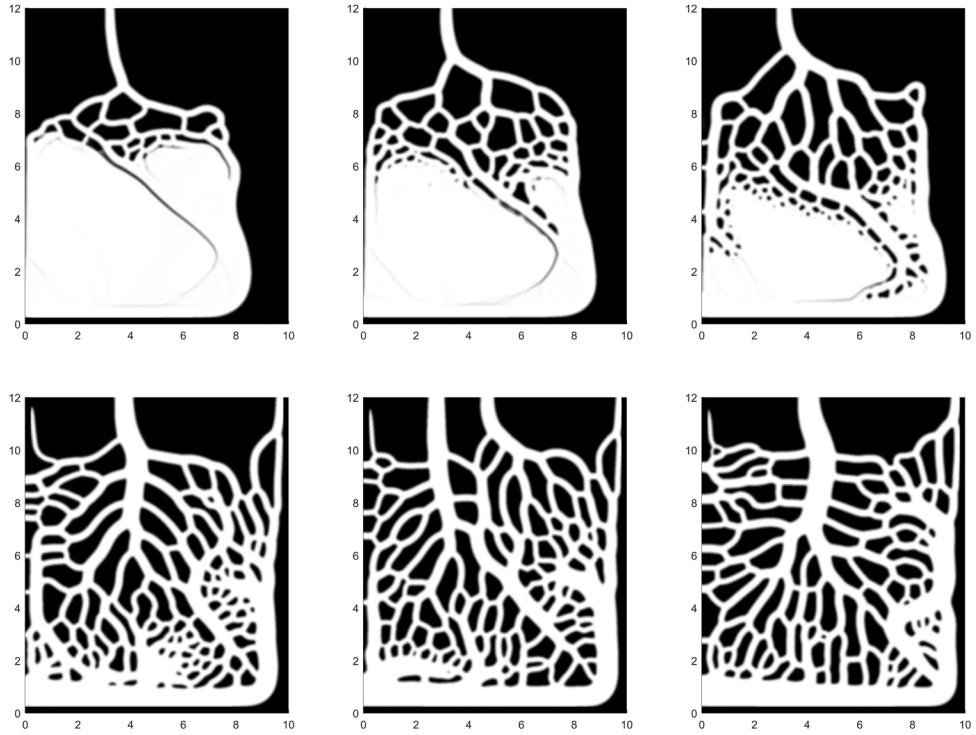


FIGURE 8 Compromise designs for different weights (w_1, w_2). Top row (from left to right): (0.8,0.2), (0.7, 0.3), and (0.5,0.5). Bottom row: (0.3,0.7), (0.2, 0.8), and (0.1,0.9).

and the goal is to limit the thermal compliance from the convection load. The mesh consists of $140 \times 21 \times 168$ FEs for both flow and temperature, the filter radius is set to 1.5 mm, and symmetry around the plane $y = 1.5$ cm is utilized to reduce the number of flow and temperature degrees of freedom to a total of around 1.7 million. The computational time was dominated by the solution of the flow state problem, which took around 80 s to solve once on a computer equipped with two 16-core Intel Xeon Gold 6130 CPUs and 384 GB RAM, making for a total of around 15 h to complete 600 MMA iterations. The thermal state and adjoint problems were solved using GMRES with an algebraic multi-grid preconditioner implemented the code AMGCL [13]. Looking at Figure 7, one may note the similarity with the 2D designs in Figures 3–6, suggesting that the set-up for the 2D problems in Figure 1 is fairly reasonable. A design-feature not captured in 2D, however, is that the larger channels in, for example, the top right and lower left corner are centered around the symmetry plane ($y = 1.5$ cm) with branches out in the y-direction into a network of smaller channels closer to the outer shell ($y \approx 3$ cm).

5.3 | The complete problem

We now present some results from the complete problem (29) with different values of the weights w_1 and w_2. Since we use continuation on $\overline{\alpha}$ and q in the fluid problem, there is some freedom in choosing the designs for the Utopia (best) and Nadir (worst) points in the scaled objectives in (21). For the flow problem, the Utopia points are obtained for the largest values of these parameters, whereas for the temperature problem, it is obtained for the lowest values. For the Nadir point, it is the other way around.

To ensure high-contrast designs for all choices of weights, with as little intermediate densities as allowed by the density filter, we used five continuation steps (this was more important for the designs with higher weight w_1 on the flow objective) with 250 iterations at each step. The Brinkman parameters were initialized to $(\overline{\alpha}, q) = (10^8, 0.01)$ and then increased to $(10^8, 0.1)$, $(10^9, 0.1)$, $(10^{10}, 0.1)$, and $(10^{10}, 0.2)$ in the respective step. To stabilize the optimization process for the larger values, the MMA parameters `asyinit`, `asyincr`, and `asydecr` were changed from initial, default values (0.5,1.2,0.7) to (0.1,1.1,0.1) at continuation step 4. This was done for all designs, but in practice, it was only necessary for the problems with larger weight on the fluid compliance, that is, $w_1 > 0.5$.

Figure 8 shows six different designs, achieving different level of compromise between minimization of the fluid compliance and the mean temperature in the solid. In the top row, where the emphasis is on minimizing the fluid compliance,

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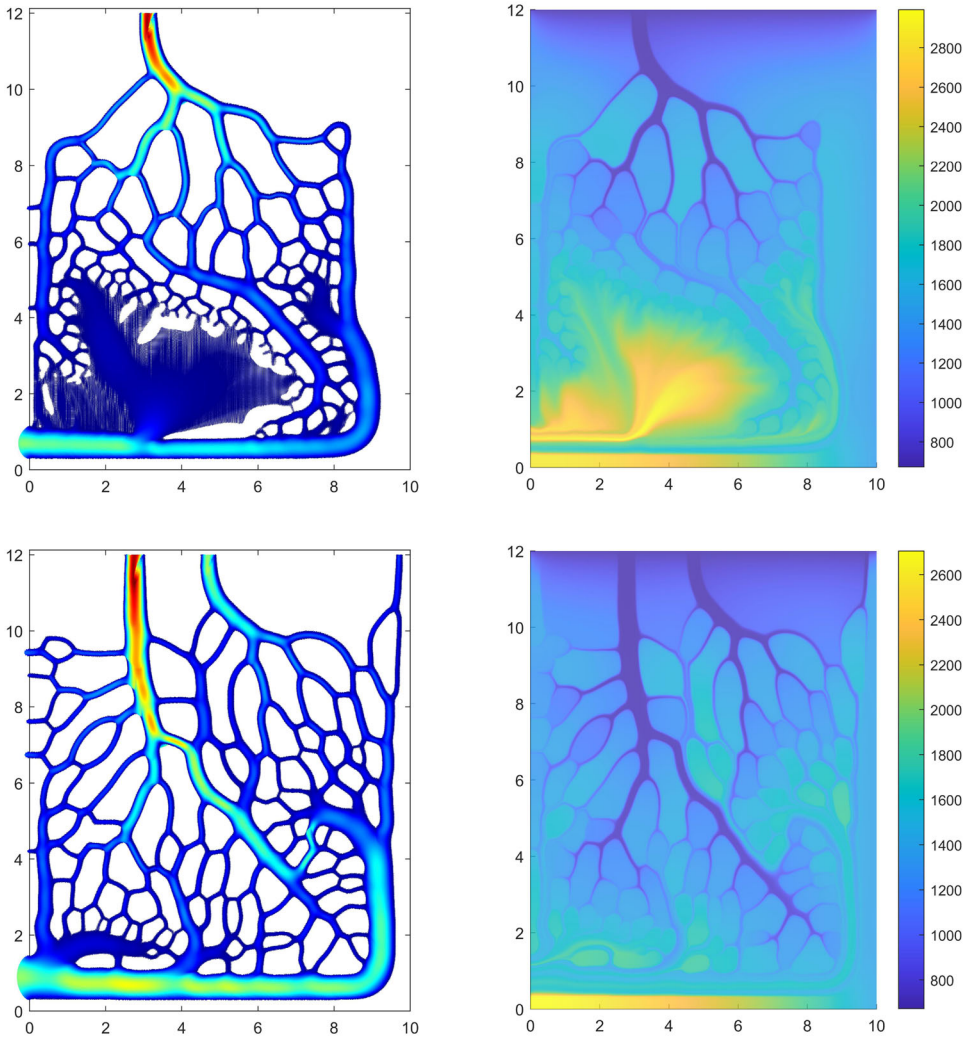


FIGURE 9 Flow and temperature fields in the designs from Figure 8 corresponding to weights (0.5,0.5) (top row) and (0.2,0.8) (bottom row)

there is a single, rather narrow inlet channel and cooling is mainly achieved by conduction in the solid. In the bottom row, where the emphasis is on minimizing the temperature, there is a big main inlet channel accompanied by two or three smaller inlets. Figure 9 shows flow and temperature in the designs corresponding to weights (0.5,0.5) and (0.2,0.8), respectively. Figure 10, finally, shows solutions to the same problem but with half the filter radius, thus allowing for appearance of much smaller features in the designs. In practise, a lower limit is imposed on the diameter of the cooling channels for manufacturability and to ensure that soot or dust particles do not get stuck in the channel; for design of gas turbine parts, this limit would be in the order of 1 mm or slightly less. Measuring for example in the lower right design in Figure 10, for weight (0.1,0.9), we find that the smallest channels would be in this range (if the design was also extruded a millimeter in the out-of-plane direction).

6 | CONCLUDING REMARKS

We have described a TO-based method for design of cooling systems in which one seeks a compromise between low temperatures and limited use of a cooling fluid. The idea and performance of the method were demonstrated numerically in both 2D and 3D for a problem-setting inspired by a real-world industrial design problem. Designs of the type seen in the bottom rows of Figures 8 and 10, with fairly uniform cooling achieved by a network of small channels resulting in a not-too-large usage of cooling medium, are probably most desirable in practice. Possible extensions of the presented work include the use of more advanced flow models and further parallelization to achieve higher resolution in 3D.

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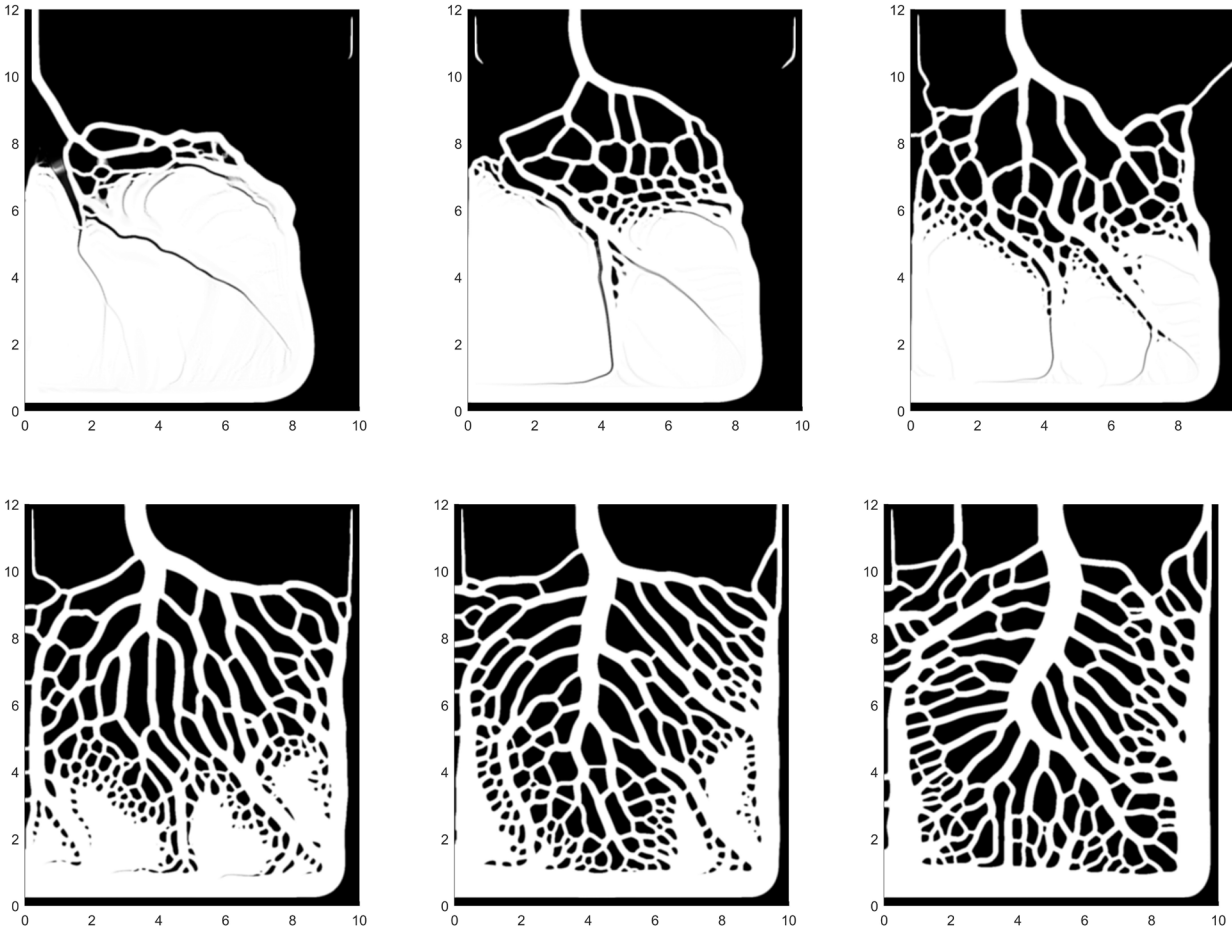


FIGURE 10 Same setup as in Figure 8 but with a smaller filter radius.

6.1 | Replication of results

A complete description of the mathematical model together with numerical values for all model parameters is provided in the paper.

ACKNOWLEDGMENTS

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APPENDIX A: EXISTENCE OF SOLUTIONS

To prove Theorem A.2, asserting the existence of a globally optimal solution to problem (22), we make use of the following lemma:

Lemma A.1. *Let* $\rho_n \in \mathcal{H} \overset{*}{\rightharpoonup} \rho \in \mathcal{H}$ *in* $L^\infty(\Omega)$, $\boldsymbol{u}_n \rightharpoonup \boldsymbol{u}$ *in* $\boldsymbol{H}^1(\Omega)$ *and* $T_n \rightharpoonup T$ *in* $H^1(\Omega)$. *Then, for a given* $(\tilde{T}, \boldsymbol{v})$ *in* $H^1(\Omega) \times \boldsymbol{H}^1(\Omega)$, *there is a subsequence such that*

$$a_T(\rho_{n'}, \boldsymbol{u}_{n'}; T_{n'}, \tilde{T}) \to a_T(\rho, \boldsymbol{u}; T, \tilde{T}) \;\; \text{and} \;\; a_F(\rho_{n'}; \boldsymbol{u}_{n'}, \boldsymbol{v}) \to a_F(\rho, \boldsymbol{u}, \boldsymbol{v}). \tag{A.1}$$

Proof. We first recall the definition (1) of the filtered design and note that the kernel $\Psi(\boldsymbol{x}, \cdot) \in L^1(\Omega)$ since this function is continuous on the bounded domain Ω. It then follows immediately from the definition of weak* convergence in $L^\infty(\Omega)$ that

$$\tilde{\rho}(\rho)(\boldsymbol{x}) - \tilde{\rho}(\rho_n)(\boldsymbol{x}) = \int_\Omega [\rho(\boldsymbol{y}) - \rho_n(\boldsymbol{y})]\Psi(\boldsymbol{x}, \boldsymbol{y})\, \mathrm{d}\boldsymbol{y} \to 0 \tag{A.2}$$

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for every $\boldsymbol{x} \in \Omega$ (not just a.e. because $\tilde{\rho}$ is actually continuous). The remainder of the proof is similar to Lemma 1 in Thore et al. [24] and thus not repeated here. □

Theorem A.2. *The design problem (22) admits at least one globally optimal solution*

Proof. Let $\{\rho_n\}$ be an infimizing sequence. Since $\mathcal{H}$ is weakly* sequentially compact, we can extract a subsequence $\{\rho_n\}$ converging weakly* to some $\rho \in \mathcal{H}$. For each ρ_n, let $\boldsymbol{u}_n = \boldsymbol{u}(\rho_n)$ and $T_n = T(\rho_n, \boldsymbol{u}_n)$ be the corresponding solution to (15). Since the bilinear form $a_F(\rho, \cdot, \cdot)$ is uniformly coercive on $\boldsymbol{V}$ and since ℓ is bounded on this space, we get $||\boldsymbol{u}_n||_{\boldsymbol{H}^1(\Omega)} \leq C$ for all n and some $C > 0$. Recalling (14), $a_T(\rho, \boldsymbol{u}; \cdot, \cdot)$ is, for a sufficiently large λ, coercive on $\mathcal{V}$ with a constant independent of both ρ and $\boldsymbol{u}$, and using $||\boldsymbol{u}_n||_{\boldsymbol{H}^1(\Omega)} \leq C$, the load-term $\ell_0(\tilde{T}) - a_T(\rho, \boldsymbol{u}; \hat{T}_0, \tilde{T})$ can be bounded on $\mathcal{V}$ independently of ρ and $\boldsymbol{u}$, so that $||T_n||_{H^1(\Omega)} \leq C_1$ for all n and some $C_1 > 0$. The sequence of velocities and temperatures is thus bounded, whence we may by the closedness of $\boldsymbol{V}$ and $\mathcal{V}$ and Lemma A.1 (and the continuity of ℓ and ℓ_0) extract a subsequence converging weakly*×weakly×weakly to some $(\rho, \boldsymbol{u}, T) \in \mathcal{H} \times \boldsymbol{V} \times \mathcal{V}$ with $(\boldsymbol{u}, T)$ solving the state problem. The proof is concluded by recalling that ρ_n was an infimizing sequence and noting that $\phi_F = \phi_F(\boldsymbol{u})$ and $\phi_T = \phi_T(\rho, T)$, defined in Equations (16) and (17), are weakly and weakly* × weakly continuous on $\boldsymbol{V}$ and $\mathcal{V}$, respectively (c.f. Lemma 6). □

APPENDIX B: POINT-WISE CONVERGENCE OF THE DISCRETIZED FILTERED DESIGN

Lemma B.1. *Consider $\tilde{\rho}_h$ defined by its elemental values in (23), and let $\{\rho_h\}$, $\rho_h \in \mathcal{H}_h$, be a sequence converging weakly* in $L^\infty(\Omega)$ to some $\rho \in \mathcal{H}$. Then*

$$\tilde{\rho}_h(\rho_h) \to \tilde{\rho}(\rho) \text{ a.e. in } \Omega. \tag{B.1}$$

Proof. Let $\varepsilon > 0$ be arbitrary. Using Equation (A.2), we find that for every h small enough,

$$|\tilde{\rho}_h(\rho_h) - \tilde{\rho}(\rho)| \leq |\tilde{\rho}(\rho_h) - \tilde{\rho}(\rho)| + |\tilde{\rho}_h(\rho_h) - \tilde{\rho}(\rho_h)| < \varepsilon/2 + |\tilde{\rho}_h(\rho_h) - \tilde{\rho}(\rho_h)| \tag{B.2}$$

for every $\boldsymbol{x} \in \cup_{e=1}^m \Omega_e$. To handle the last term, we first note that, since the elements are disjoint, $\boldsymbol{x} \in \Omega_e$ for exactly one e. Let $\mathcal{N}_e$ denote the set of elements whose centroid is within (Euclidean) distance R of $\boldsymbol{x}_e$. Making use of the integral mean value theorem in the last line we get

$$\begin{aligned} r &\equiv \left| \sum_{f \in \mathcal{N}_e} \rho_f V_f \Psi(\boldsymbol{x}_e, \boldsymbol{x}_f) - \int_\Omega \rho_h(\boldsymbol{y}) \Psi(\boldsymbol{x}, \boldsymbol{y}) \, \mathrm{d}\boldsymbol{y} \right| \\ &= \left| \sum_{f \in \mathcal{N}_e} \rho_f V_f \Psi(\boldsymbol{x}_e, \boldsymbol{x}_f) - \sum_{f=1}^m \rho_f \int_{\Omega_f} \Psi(\boldsymbol{x}, \boldsymbol{y}) \, \mathrm{d}\boldsymbol{y} \right| \\ &\leq \sum_{f \in \mathcal{N}_e} \rho_f V_f |\Psi(\boldsymbol{x}_e, \boldsymbol{x}_f) - \Psi(\boldsymbol{x}, \boldsymbol{\xi}_f)| + \sum_{f \in \{1,\dots,m\} \setminus \mathcal{N}_e} \rho_f V_f |\Psi(\boldsymbol{x}, \boldsymbol{\xi}_f)|, \end{aligned} \tag{B.3}$$

for some $\boldsymbol{\xi}_f \in \Omega_f$. The first term in the last line comes from those elements whose centroid lies within the support of the filter, and the other from those whose centroid lies outside. The centroid being located inside the element, the distance between any point in an element and its centroid will tend to zero as $h \to 0$. So, for small enough h, the continuity of Ψ gives $|\Psi(\boldsymbol{x}, \boldsymbol{\xi}_f) - \Psi(\boldsymbol{x}_e, \boldsymbol{x}_f)| < \varepsilon/(2|\mathcal{N}_e|V_f)$ for $f \in \mathcal{N}_e$ and $|\Psi(\boldsymbol{x}, \boldsymbol{\xi}_f)| < \varepsilon/(2(m - |\mathcal{N}_e|)V_f)$ for $f \notin \mathcal{N}_e$ (here $|\mathcal{N}_e|$ denotes the number of elements in $\mathcal{N}_e$). Substituting this into (B.3) and noting that $\rho_f \leq 1$ leads to

$$r < \sum_{f \in \mathcal{N}_e} \frac{\varepsilon V_f}{2|\mathcal{N}_e|V_f} + \sum_{f \in \{1,\dots,m\} \setminus \mathcal{N}_e} \frac{\varepsilon V_f}{2(m - |\mathcal{N}_e|)V_f} = \varepsilon. \tag{B.4}$$

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Based on this result and the fact that every $\boldsymbol{x} \in \cup_{e=1}^{m} \Omega_e$ belongs to some Ω_e, we conclude that

$$\tilde{\rho}_h(\rho_h)(\boldsymbol{x}) = \frac{\sum_{f \in \mathcal{N}_e} \rho_f V_f \Psi(\boldsymbol{x}_e, \boldsymbol{x}_f)}{\sum_{f \in \mathcal{N}_e} V_f \Psi(\boldsymbol{x}_e, \boldsymbol{x}_f)} \to \frac{\int_\Omega \rho_h(\boldsymbol{y}) \Psi(\boldsymbol{x}, \boldsymbol{y}) d\boldsymbol{y}}{1} = \tilde{\rho}(\rho_h)(\boldsymbol{x}) \quad \text{a.e. in } \Omega, \tag{B.5}$$

using the fact that

$$\int_\Omega \Psi(\boldsymbol{x}, \boldsymbol{y}) d\boldsymbol{y} = 1 \quad \forall \boldsymbol{x} \in \Omega \tag{B.6}$$

with the kernel chosen as in (2). Now Equation (B.1) follows from (B.2). □

APPENDIX C: DERIVATIVES

The matrix problem corresponding to (24) reads: Find $(\boldsymbol{u}, \boldsymbol{r}_F) \in \mathbb{R}^{n_F + n_{F0}}$ such that

$$\begin{aligned} \boldsymbol{K}_F \boldsymbol{u} &= \boldsymbol{f} + \boldsymbol{C}_F^\top \boldsymbol{r}_F \\ \boldsymbol{C}_F \boldsymbol{u} &= \boldsymbol{0} \end{aligned} \tag{C.1}$$

where $\boldsymbol{r}_F$ contains reaction loads (Lagrange multipliers) arising due to the constraint $\boldsymbol{C}_F \boldsymbol{u} = \boldsymbol{0}$. The derivative of the negative compliance with respect to a filtered variable ρ_e can be obtained as follows: First we introduce the augmented functional

$$\hat{\phi}_F = -\boldsymbol{f}^\top \boldsymbol{u} + \boldsymbol{\lambda}_1^\top (\boldsymbol{K}_F \boldsymbol{u} - \boldsymbol{f} - \boldsymbol{C}_F^\top \boldsymbol{r}_F) + \boldsymbol{\lambda}_2^\top \boldsymbol{C}_F \boldsymbol{u}, \tag{C.2}$$

where $\boldsymbol{\lambda}_1$ and $\boldsymbol{\lambda}_2$ are arbitrary vectors. Since (C.1) is satisfied, the values of $\hat{\phi}_F$ and ϕ_F are identical everywhere, so their derivatives coincide. Now, making use of (C.1) to eliminate the derivatives of $\boldsymbol{\lambda}_1$ and $\boldsymbol{\lambda}_2$,

$$\begin{aligned} \frac{\partial \hat{\phi}_F}{\partial \rho_e} &= -\boldsymbol{f}^\top \frac{\partial \boldsymbol{u}}{\partial \rho_e} + \boldsymbol{\lambda}_1^\top \left(\frac{\partial \boldsymbol{K}_F}{\partial \rho_e} \boldsymbol{u} + \boldsymbol{K}_F \frac{\partial \boldsymbol{u}}{\partial \rho_e} - \boldsymbol{f} - \boldsymbol{C}_F^\top \frac{\partial \boldsymbol{r}_F}{\partial \rho} \right) + \boldsymbol{\lambda}_2^\top \boldsymbol{C}_F \frac{\partial \boldsymbol{u}}{\partial \rho_e} \\ &= \boldsymbol{\lambda}_1^\top \frac{\partial \boldsymbol{K}_F}{\partial \rho_e} \boldsymbol{u} + \left(-\boldsymbol{f}^\top + \boldsymbol{\lambda}_1^\top \boldsymbol{K}_F + \boldsymbol{\lambda}_2^\top \boldsymbol{C}_F \right) \frac{\partial \boldsymbol{u}}{\partial \rho_e} - \boldsymbol{\lambda}_1 \boldsymbol{C}_F^\top \frac{\partial \boldsymbol{r}_F}{\partial \rho_e}. \end{aligned} \tag{C.3}$$

Here the complicating terms $\frac{\partial \boldsymbol{u}}{\partial \rho_e}$ and $\frac{\partial \boldsymbol{r}_F}{\partial \rho_e}$ vanish by choosing $\boldsymbol{\lambda}_1$ and $\boldsymbol{\lambda}_2$ to satisfy

$$\begin{aligned} \boldsymbol{K}_F \boldsymbol{\lambda}_1 &= \boldsymbol{f} - \boldsymbol{C}_F^\top \boldsymbol{\lambda}_2 \\ \boldsymbol{C}_F \boldsymbol{\lambda}_1 &= \boldsymbol{0}. \end{aligned} \tag{C.4}$$

This system is similar in structure to (C.1) so we may identify $\boldsymbol{\lambda}_1$ with $\boldsymbol{u}$, whence the final expression for the derivative becomes

$$\frac{\partial \phi_F}{\partial \rho_e} = \boldsymbol{u}^\top \frac{\partial \boldsymbol{K}_F}{\partial \rho_e} \boldsymbol{u}. \tag{C.5}$$

The matrix problem corresponding to (25) reads: Find $(\boldsymbol{T}, \boldsymbol{r}_T) \in \mathbb{R}^{n + n_{T0}}$ such that

$$\begin{aligned} \boldsymbol{K}_T \boldsymbol{T} &= \boldsymbol{f}_T + \boldsymbol{C}_T^\top \boldsymbol{r}_T \\ \boldsymbol{C}_T \boldsymbol{T} &= \boldsymbol{T}_0. \end{aligned} \tag{C.6}$$

To get the derivative of (32) with respective to optimization variable ρ_e, we consider the augmented functional

$$\hat{\phi}_T = \boldsymbol{L}^\top \boldsymbol{T} + \boldsymbol{\lambda}_T^\top (\boldsymbol{K}_T \boldsymbol{T} - \boldsymbol{f}_T - \boldsymbol{C}_T^\top \boldsymbol{r}_T) + \boldsymbol{\lambda}_3^\top (\boldsymbol{C}_T \boldsymbol{T} - \boldsymbol{T}_0) + \boldsymbol{\lambda}_F^\top (\boldsymbol{K}_F \boldsymbol{u} - \boldsymbol{f} - \boldsymbol{C}_F^\top \boldsymbol{r}_F) + \boldsymbol{\lambda}_4^\top \boldsymbol{C}_F \boldsymbol{u} \tag{C.7}$$

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where the $\boldsymbol{\lambda}$s are arbitrary. Requiring that $\boldsymbol{C}_T\boldsymbol{\lambda}_3 = \boldsymbol{0}$ and $\boldsymbol{C}_F\boldsymbol{\lambda}_4 = \boldsymbol{0}$, we can get rid of derivatives with respect to the thermal and fluid reaction loads directly and get

$$\begin{aligned}\frac{\partial\hat{\phi}_T}{\partial\rho_e} &= \frac{\partial \boldsymbol{L}}{\partial\rho_e}^{\mathsf{T}}\boldsymbol{T} + \boldsymbol{L}^{\mathsf{T}}\frac{\partial\boldsymbol{T}}{\partial\rho_e} + \boldsymbol{\lambda}_T^{\mathsf{T}}\left(\frac{\partial\boldsymbol{K}_T}{\partial\rho_e}\boldsymbol{T} + \boldsymbol{K}_T\frac{\partial\boldsymbol{T}}{\partial\rho_e} - \frac{\partial\boldsymbol{f}_T}{\partial\rho_e}\right) + \boldsymbol{\lambda}_2^{\mathsf{T}}\boldsymbol{C}_T\frac{\partial\boldsymbol{T}}{\partial\rho_e} + \boldsymbol{\lambda}_F^{\mathsf{T}}\left(\frac{\partial\boldsymbol{K}_F}{\partial\rho_e}\boldsymbol{u} + \boldsymbol{K}_F\frac{\partial\boldsymbol{u}}{\partial\rho_e}\right) + \boldsymbol{\lambda}_4^{\mathsf{T}}\boldsymbol{C}_F\frac{\partial\boldsymbol{u}}{\partial\rho_e}\\ &= \frac{\partial \boldsymbol{L}}{\partial\rho_e}^{\mathsf{T}}\boldsymbol{T} + \boldsymbol{\lambda}_T^{\mathsf{T}}\frac{\partial\boldsymbol{K}_T}{\partial\rho_e}\boldsymbol{T} + \boldsymbol{\lambda}_F^{\mathsf{T}}\frac{\partial\boldsymbol{K}_F}{\partial\rho_e}\boldsymbol{u} + \left(\boldsymbol{L}^{\mathsf{T}} + \boldsymbol{\lambda}_T^{\mathsf{T}}\boldsymbol{K}_T + \boldsymbol{\lambda}_2^{\mathsf{T}}\boldsymbol{C}_T\right)\frac{\partial\boldsymbol{T}}{\partial\rho_e} + (\boldsymbol{\lambda}_F^{\mathsf{T}}\boldsymbol{K}_F + \boldsymbol{C}_F^{\mathsf{T}}\boldsymbol{\lambda}_4)\frac{\partial\boldsymbol{u}}{\partial\rho_e}.\end{aligned} \tag{C.8}$$

If we require that $\boldsymbol{\lambda}_T$ and $\boldsymbol{\lambda}_2$ satisfy

$$\begin{aligned}\boldsymbol{K}_T^{\mathsf{T}}\boldsymbol{\lambda}_T &= -\boldsymbol{L} - \boldsymbol{C}_T^{\mathsf{T}}\boldsymbol{\lambda}_2\\ \boldsymbol{C}_T\boldsymbol{\lambda}_T &= \boldsymbol{0}\end{aligned} \tag{C.9}$$

we get

$$\frac{\partial\phi_T}{\partial\rho_e} = \frac{\partial \boldsymbol{L}}{\partial\rho_e}^{\mathsf{T}}\boldsymbol{T} + \boldsymbol{\lambda}_T^{\mathsf{T}}\frac{\partial\boldsymbol{K}_T}{\partial\rho_e}\boldsymbol{T} - \boldsymbol{\lambda}_T^{\mathsf{T}}\frac{\partial\boldsymbol{f}_T}{\partial\rho_e} + \boldsymbol{\lambda}_F^{\mathsf{T}}\frac{\partial\boldsymbol{K}_F}{\partial\rho_e}\boldsymbol{u} + (\boldsymbol{\lambda}_F^{\mathsf{T}}\boldsymbol{K}_F + \boldsymbol{C}_F^{\mathsf{T}}\boldsymbol{\lambda}_4)\frac{\partial\boldsymbol{u}}{\partial\rho_e}. \tag{C.10}$$

Here $\boldsymbol{K}_T = \boldsymbol{K}_T(\boldsymbol{\rho}, \boldsymbol{u}(\boldsymbol{\rho}))$ and $\boldsymbol{f}_T = \boldsymbol{f}_T(\boldsymbol{\rho}, \boldsymbol{u}(\boldsymbol{\rho}))$, so, with a slight abuse of notation and letting $\boldsymbol{e}_i$ denote a standard basis vector in $\mathbb{R}^{n_f}$,

$$\begin{aligned}\frac{\partial\phi_T}{\partial\rho_e} &= \frac{\partial \boldsymbol{L}}{\partial\rho_e}^{\mathsf{T}}\boldsymbol{T} + \boldsymbol{\lambda}_T^{\mathsf{T}}\frac{\partial\boldsymbol{K}_T}{\partial\rho_e}\boldsymbol{T} - \boldsymbol{\lambda}_T^{\mathsf{T}}\frac{\partial\boldsymbol{f}_T}{\partial\rho_e} - \boldsymbol{\lambda}_T^{\mathsf{T}}\sum_{i=1}^{n_F}\frac{\partial\boldsymbol{f}_T}{\partial u_i}\boldsymbol{e}_i^{\mathsf{T}}\frac{\partial\boldsymbol{u}}{\partial\rho_e} + \boldsymbol{\lambda}_F^{\mathsf{T}}\frac{\partial\boldsymbol{K}_F}{\partial\rho_e}\boldsymbol{u} + \boldsymbol{\lambda}_T^{\mathsf{T}}\sum_{i=1}^{n_F}\frac{\partial\boldsymbol{K}_T}{\partial u_i}\boldsymbol{T}\boldsymbol{e}_i^{\mathsf{T}}\frac{\partial\boldsymbol{u}}{\partial\rho_e}\\ &\quad + (\boldsymbol{\lambda}_F^{\mathsf{T}}\boldsymbol{K}_F + \boldsymbol{C}_F^{\mathsf{T}}\boldsymbol{\lambda}_4)\frac{\partial\boldsymbol{u}}{\partial\rho_e} = \frac{\partial \boldsymbol{L}}{\partial\rho_e}^{\mathsf{T}}\boldsymbol{T} + \boldsymbol{\lambda}_T^{\mathsf{T}}\frac{\partial\boldsymbol{K}_T}{\partial\rho_e}\boldsymbol{T} - \boldsymbol{\lambda}_T^{\mathsf{T}}\frac{\partial\boldsymbol{f}_T}{\partial\rho_e} + \boldsymbol{\lambda}_F^{\mathsf{T}}\frac{\partial\boldsymbol{K}_F}{\partial\rho_e}\boldsymbol{u}\end{aligned} \tag{C.11}$$

if $\boldsymbol{\lambda}_F$ and $\boldsymbol{\lambda}_4$ solve

$$\begin{aligned}\boldsymbol{K}_F\boldsymbol{\lambda}_F &= \sum_{i=1}^{n_F}\left(-\boldsymbol{\lambda}_T^{\mathsf{T}}\frac{\partial\boldsymbol{K}_T}{\partial u_i}\boldsymbol{T} + \boldsymbol{\lambda}_T^{\mathsf{T}}\frac{\partial\boldsymbol{f}_T}{\partial u_i}\right)\boldsymbol{e}_i + \boldsymbol{C}_F^{\mathsf{T}}\boldsymbol{\lambda}_4\\ \boldsymbol{C}_F\boldsymbol{\lambda}_F &= \boldsymbol{0}.\end{aligned} \tag{C.12}$$

As for the right-hand side in the first of these (adjoint) equations, one may choose to assemble it in an element-oriented way by noting that

$$\sum_{i=1}^{n_F}\boldsymbol{\lambda}_T^{\mathsf{T}}\frac{\partial\boldsymbol{K}_T}{\partial u_i}\boldsymbol{T}\boldsymbol{e}_i = \sum_{i=1}^{n_F}\boldsymbol{\lambda}_T^{\mathsf{T}}\left(\sum_{e=1}^{m}\frac{\partial\boldsymbol{K}_{Te}}{\partial u_i}\right)\boldsymbol{T}\boldsymbol{e}_i = \sum_{e=1}^{m}\left(\sum_{i\in\mathcal{N}_e}\boldsymbol{\lambda}_{Te}^{\mathsf{T}}\frac{\partial\boldsymbol{K}_{Te}}{\partial u_i}\boldsymbol{T}_e\boldsymbol{e}_i\right), \tag{C.13}$$

where $\mathcal{N}_e$ contains the velocity degrees of freedom of element e. Element-oriented assembly saves some computations compared to the node-oriented assembly, but while is trivial to parallelize, the element-oriented assembly may lead to race-conditions if two elements try to contribute to the same entry i at the same time. However, the computational time for solving problem (22) is typically completely dominated by the time for solving the state (and adjoint) problems. For simplicity, we have, therefore, implemented an element-oriented assembly procedure using the OpenMP atomic directive to avoid race conditions.

The temperature stiffness matrix $\boldsymbol{K}_{Te}$ in (C.13) can be divided into one matrix associated with conduction and convection and one with stabilization. For the latter we get (letting $\boldsymbol{B}_e = \nabla\boldsymbol{N}_e$)

$$\begin{aligned}\frac{\partial\boldsymbol{K}_{T\tau e}}{\partial u_i} &= \int_{\Omega_e}\left[\tau\frac{\partial}{\partial u_i}(c\boldsymbol{B}_e^{\mathsf{T}}\boldsymbol{u}_h)(c\boldsymbol{u}_h^{\mathsf{T}}\boldsymbol{B}_e) + \frac{\partial\tau}{\partial u_i}(c\nabla\boldsymbol{B}_e^{\mathsf{T}}\boldsymbol{u}_h)(c\boldsymbol{u}_h^{\mathsf{T}}\boldsymbol{B}_e)\right]\mathrm{d}V\\ &= \int_{\Omega_e}\left[\tau(c\boldsymbol{B}_e^{\mathsf{T}}\boldsymbol{N}_{e\cdot j}^F)(c\boldsymbol{u}_h^{\mathsf{T}}\boldsymbol{B}_e) + \tau(c\boldsymbol{B}_e^{\mathsf{T}}\boldsymbol{u}_h)(c(\boldsymbol{N}_{e\cdot j}^F)^{\mathsf{T}}\boldsymbol{B}_e) + \frac{\partial\tau}{\partial u_i}(c\boldsymbol{B}_e^{\mathsf{T}}\boldsymbol{u}_h)(c\boldsymbol{u}_h^{\mathsf{T}}\boldsymbol{B}_e)\right]\mathrm{d}V,\end{aligned} \tag{C.14}$$

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where $\boldsymbol{N}^F_{e\cdot j}$ denotes the jth column of the fluid shape function matrix and j is the local number of the ith velocity component. Here (specializing to 2D; the 3D case is analogous)

$$\frac{\partial \tau}{\partial u_i} = \frac{h}{2}\left(\frac{\partial \overline{x}}{\partial u_i}u_x + \overline{x}\frac{\partial u_x}{\partial u_i} + \frac{\partial \overline{y}}{\partial u_i}u_y + \overline{y}\frac{\partial u_y}{\partial u_i}\right)\frac{1}{||\boldsymbol{u}_h||^2} + \frac{h}{2}(\overline{x}u_x + \overline{y}u_y)\frac{\partial}{\partial u_i}\left(\frac{1}{||\boldsymbol{u}_h||^2}\right), \tag{C.15}$$

where

$$\frac{\partial \overline{x}}{\partial u_i} = \frac{\partial \overline{x}}{\partial Pe_x}\frac{\partial Pe_x}{\partial u_i} = \left(1 - \coth^2 Pe_x + 1/Pe_x^2\right)\frac{\partial u_x}{\partial u_i}\frac{h}{k}, \tag{C.16}$$

in which (assuming the usual x-y-x-y-ordering of the degrees of freedom)

$$\frac{\partial u_x}{\partial u_i} = \begin{cases} N_{(i+1)/2} & \text{if } i \text{ is odd} \\ 0 & \text{otherwise} \end{cases} \tag{C.17}$$

where $N_{(i+1)/2}$ is a shape function. The last term in (C.15) is given by

$$\frac{\partial}{\partial u_i}\left(\frac{1}{||\boldsymbol{u}_h||^2}\right) = \frac{-2}{||\boldsymbol{u}_h||^4}\boldsymbol{u}_h^{\mathsf{T}}\frac{\partial \boldsymbol{u}_h}{\partial u_i}. \tag{C.18}$$

To compute the two last terms in (C.11), we also need to account for the explicit design-dependence of the stabilization parameter. To this end, we note that

$$\boldsymbol{\lambda}_{\mathrm{T}e}^{\mathsf{T}}\frac{\partial \boldsymbol{K}_{Te}}{\partial \rho_e}\boldsymbol{T}_e = \boldsymbol{\lambda}_{\mathrm{T}}^{\mathsf{T}}\int_{\Omega_e}\frac{\partial \tau}{\partial \rho_e}(\boldsymbol{B}_e^{\mathsf{T}}\boldsymbol{u}_h)(\boldsymbol{u}_h^{\mathsf{T}}\boldsymbol{B}_e)\,\mathrm{d}V\boldsymbol{T}_e \tag{C.19}$$

where

$$\frac{\partial \tau}{\partial \rho_e} = \frac{\partial \tau}{\partial Pe_x}\frac{\partial Pe_x}{\partial \rho_e} + \frac{\partial \tau}{\partial Pe_y}\frac{\partial Pe_y}{\partial \rho_e}, \tag{C.20}$$

in which, recalling (28),

$$\frac{\partial \tau}{\partial Pe_x} = \frac{h}{2c}\frac{\partial \overline{x}}{\partial Pe_x}\frac{1}{||\boldsymbol{u}_h||^2} = \frac{h}{2c}\left(1 - \coth^2 Pe_x + \frac{1}{Pe_x^2}\right)\frac{1}{||\boldsymbol{u}_h||^2} \quad \text{and} \quad \frac{\partial Pe_x}{\partial \rho_e} = \frac{-u_x h}{k^2}\frac{\partial k}{\partial \rho_e}. \tag{C.21}$$

The load-vector contribution from the stabilized source term in (26) can be written as

$$\sum_{e=1}^{m}\int_{\Omega_e}\tau(c\boldsymbol{u}_h\cdot\nabla\tilde{T}_h)Q\,\mathrm{d}V = \sum_{e=1}^{m}\underbrace{\int_{\Omega_e}\tau(c\boldsymbol{u}_h^{\mathsf{T}}\boldsymbol{B}_e)Q\,\mathrm{d}V}_{\boldsymbol{f}_e^{\mathsf{T}}}\boldsymbol{C}_e\tilde{\boldsymbol{T}}, \tag{C.22}$$

where $\boldsymbol{f}_e$ is the contribution from element e. The derivatives of $\boldsymbol{f}_e$ are given by

$$\frac{\partial \boldsymbol{f}_e}{\partial u_i} = \int_{\Omega_e} c\left(\frac{\partial \tau}{\partial u_i}\boldsymbol{B}_e^{\mathsf{T}}\boldsymbol{u}_h + \tau\nabla\boldsymbol{N}_e^{\mathsf{T}}\frac{\partial \boldsymbol{u}_h}{\partial u_i}\right)Q\,\mathrm{d}V \text{ and } \frac{\partial \boldsymbol{f}_e}{\partial \rho_e} = \int_{\Omega_e}\frac{\partial \tau}{\partial \rho_e}(c\boldsymbol{B}_e^{\mathsf{T}}\boldsymbol{u}_h)Q\,\mathrm{d}V. \tag{C.23}$$

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